

The New Experts:
Reproducing and Recasting the Cultural Imaginaries of Biomedical Expertise during COVID-19
on the Joe Rogan Podcast

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Honors Thesis
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April 10th, 2026

Acknowledgements

So many people were integral to the completion of this project, and I would like to take a moment to thank you all for your support. First and foremost, Dr. Weed, who has been beyond supportive of me not only in this thesis but for the past four years as my undergrad journey unfolded— you have absolutely shaped me both academically and personally, and I will always be so incredibly thankful to the guidance you have given me. I would also like to thank Dr. Thrailkill, whose feedback took this thesis to the next level. Equally as important are my friends and family— especially my Mother Elissa, Alex, Lyn, and Harper, who listened to my jumbled ideas for hours and encouraged me regardless, sitting with me and listening as this thesis took form. Finally, I would like to thank the Baristas at Lanzas, who saw all too much of me in their cozy establishment, and who made me wonderful matcha lattes that fueled my thesis. I couldn't have done it without you all.

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Abstract

Science is essential to everyday life, informing the way we move through the world and allowing us to make monumental decisions about our government, our safety, and our health. Perhaps no moment illustrated this more urgently than the COVID-19 pandemic, where engagement with the science of vaccines had profound consequences for public health. Yet engaging with scientific claims is incredibly difficult; as Philosopher of Science Elizabeth Anderson recognizes, laypersons cannot directly evaluate scientific claims and so are left "judg[ing] what to believe by judging whom to believe." During COVID-19, a new class of alternative-experts espoused claims about vaccine safety that ran counter to mainstream science yet commanded enormous credibility with lay audiences. One can understand their rise in our fractured epistemological landscape that, as Baier and Lyotard theorize, produces distinct epistemic spheres each with their own experts and forms of truth. Using the tools of rhetorical analysis, I draw out the rhetorical moves through which these figures generate credibility. Analyzing a case study of guests on the Joe Rogan Experience podcast who appeared in the second half of 2021, I argue that the guests generated expert credibility by reproducing and recasting the cultural imaginaries of insider biomedical expertise. Chapter One draws on Bourdieu's theory of symbolic capital to examine how alternative-experts reproduce academic credentials, embodied experience, and communal validation to construct insider status. Chapter Two argues that this insider status is recast through populist and democratic appeals, creating a dual positionality uniquely persuasive to Rogan's viewership. Ultimately, this demonstrates the rhetorical nature of expertise, and the importance of using rhetoric to understand health communication strategies moving forward.

Introduction

Who Can You Trust?

Science is essential to everyday life. It informs the way we move through the world, and allows us to make monumental decisions about our government, our safety, and our health: so engaging with scientific claims is essential. However, doing so is also incredibly difficult. As Philosopher of Science Elizabeth Anderson recognizes in her work, “Democracy, Public Policy, and Lay Assessments of Scientific Testimony,” nonscientists, or laypersons, often cannot directly evaluate the claims of scientific research, simply because they do not have scientific training. Therefore, they must rely on the assessments of others (Anderson 144). The role of the layperson, then, is “judg[ing] what to believe by judging whom to believe,” in other words, making a second-order assessment of who to trust as opposed to making a first-order assessment concerning the scientific claim at hand (145). These second-order judgments necessarily dominate scientific discourse amongst laypersons.

This argument, however, relies on the assumption that an expert will be correct about the scientific claim. While this may not always be the case, probability dictates that experts are more likely to get it right than non-experts. For this reason, it is a convenient heuristic, or short-cut that laypersons can use to form judgments quickly without having to analyze every detail. However, it creates another issue: How does one decide who is an expert? And when the experts disagree, how does one determine who to trust?

Consider this example, and decide if this figure would be worthy of trusting when it comes to COVID and COVID vaccine science:

Dr. X appeared on a popular podcast on December 31st, 2021 to talk about COVID and COVID-vaccines. At the start of the episode, the podcast host asks him to describe his

experience and relevance, and it takes Dr X. 7 and a half minutes¹ to do so: Dr. X describes himself as a physician-scientist who received an “MD PhD scholarship at Northwestern University” and who pursued his graduate bench science training at the prestigious “Salk Institute,” which is recognized in scientific circles as highly renowned center for vaccine and biology research, with a prestige he conveys to the layperson argument by stating there were no fewer than “seven Nobel laureates” there while he worked there. He highlights his singularity as a researcher not only through the prestige of the institute itself, but by highlighting his renowned skill when he states Salk is a place where “graduate students normally aren’t allowed to go.” The prestige of his training and singularity of accomplishments increases his credibility. While at Salk, Dr. X had a “series of discoveries” which formed “the basis of the RNA technology platform that [gave] rise to these [COVID-19] vaccines.” After the Salk Institute, he pursued a “fellowship at Harvard,” “ran over 100 clinical trials mostly in the vaccine space,” has “won literally billions of dollars in federal grants and contracts,” has been “involved in every major outbreak since AIDS,” and has “serv[ed] as a study section chair” for the National Institute of Health (Rogan, “Malone” [00:01:30 - 00:08:55])

Dr. X establishes his scientific expertise through his prestigious educational and prolific research credentials. These credentials, alongside his focus on vaccinology (specifically, in mRNA vaccines) and experience with previous outbreaks, should bolster his trustworthiness as a scientist, especially when it comes to pandemics and the mRNA-based COVID vaccines. The next three hours of the podcast episode furthered Dr. X’s expert status.

Despite his seeming expert scientific status, Dr. X is actually Dr Robert Malone, a physician who many condemn for spreading misinformation. In fact, during the episode, which was hosted on the Joe Rogan Experience podcast, he decried the COVID vaccine, alongside

¹ Dr. Malone calls this the “short version” of his credentials.

made claims about Ivermectin, Hydroxychloroquine and Post-COVID vaccine myocarditis which were unsupported by scientific evidence. This draws our attention to the complex nature of discerning expertise.

Indeed, this episode prompted 270 US clinical and research health professionals to condemn Dr. Robert Malone's claims on the Joe Rogan Podcast by writing a letter to Spotify asking them to take action to mitigate the spread of misinformation on the platform. ("An Open Letter to Spotify") These experts represent an expertise which adheres to mainstream treatments and scientific ideas, and thus are "mainstream-experts." While denouncing Dr. Malone's stance, these mainstream-experts called his expertise into question. Dr. Jessica Malaty Rivera, the infectious disease epidemiologist and Boston's Children's Hospital research fellow who led the efforts to write the letter, states that she was prompted to take action because she was "horrified to discover that people in her life, whom she considered to be "quite wise and discerning," were "hoodwinked by Malone's patina of academic credibility, considering his views on the vaccine legitimate" (Dickson). The fact that Rivera's "wise and discerning" peers believe him demonstrates how this isn't an issue of ignorance or foolishness. Dr. Malone possesses certain academic qualities which establish him as trustworthy and a credible expert in the eyes of certain laypersons. Thus, while many other physicians and scientists condemn Dr. Malone to a fringe position in the scientific community, many laypersons continue to perceive him as an expert, drawing to our attention the unstable categorization of expertise and making us question, when it comes to COVID and COVID vaccine discourse, who gets to claim it and why?

This thesis seeks to explore that question. I will argue that alternative-experts like Dr. Malone generate credibility with lay audiences by reproducing and recasting the cultural imaginaries of insider biomedical expertise. In this way, the perception of expertise is rhetorical,

and the potency of the persuasive appeal imbues the putative biomedical expert with their authority.

Expertise in a Time of Epistemological Fragmentation

In the fractured epistemological landscape of COVID-19 vaccine discourse, a new class of scientific voices emerged. These figures, encapsulated here by the term *alternative-expert*, espoused a variety of alternative claims concerning COVID vaccine safety, including the need for alternatives like ivermectin, which ran counter to traditional vaccine science and earned them a spot at the fringes of established scientific spheres. They are experts because they occupy a position of trust and credibility for some *laypersons*, aka nonbiomedical figures without specialist expert knowledge. These alternative-experts occupy a role of similar credibility to the *mainstream-experts*, who differ in the fact that they and their claims are recognized by established scientific and medical institutions as authoritative voices on COVID and COVID-vaccines, and espouse mainstream sentiments about the pandemic and the treatments. This broad conceptualization of expertise applies equally to anyone who commands biomedical authority, trust, respect, and is listened to, regardless of institutional standing.

This poses a seeming paradox: how can there be both alternative-experts and mainstream-experts? A philosopher would point out that these two types of expertise are incommensurate. Yet we must acknowledge these two forms of expertise— to deny them is akin to denying the impact of race. True, race is a social construct, not a biological one, but as a society we treat race as real, and thus it has real world impacts. Similarly, expertise is a social construct, so perceptions of expertise carry real consequences for health outcomes. I argue that the perception of expertise ultimately governs expert status, so we must make an effort to understand how rhetorical moves affirm individual's status as experts and thus lead to their impact on health.

Postmodernist thought serves as the best framework through which to understand this fragmentation, as it acknowledges the role of individuals in constructing a multiplicity of truths, and thus allows for the articulation of *alternative-truths* and their corollary of *alternative-experts*. Indeed, Philosopher Christian Baier draws upon the work of Lyotard, arriving at the idea that our time “reject[s] only the idea of universal truth in favor of a multiplicity of individual truths” (Baier 96). In formulating this construction, Baier predominantly relies on Lyotard’s 1979 book, *The Postmodern Condition*, which defines the postmodern as “incredulity towards metanarratives,” where metanarratives function as overarching stories humans tell ourselves about our history and goals. These metanarratives are used to legitimize our cultural practices and ways of knowledge making– for example, the metanarrative of progress, in which history continually marches towards progress and betterment (Lyotard xxiv). In the absence of cohesive, legitimizing narratives of the world, little narratives or “petit recits” proliferate. Baier rephrases this breakdown as an “epistemic fragmentation” of society, in which the “epistemic foundation” (aka, the metanarrative) has failed and has been replaced by a multitude of distinct “epistemic spheres” aka, petit recits (Baier 96). Under this construction, each epistemic sphere has its own version of truth: it can be used to explain why alternative facts proliferate; they are plain facts when viewed from another epistemic sphere. Baier applies his argument to the epistemic sphere of the MAGA movement (epitomized by so called fake news) and posits Trump’s centrality in shaping this sphere. Thus, alternative epistemic spheres not only have alternative-facts and knowledge making systems but have differing knowledge-makers or experts.

A common critique of this line of thinking is that it situates all truths on the same level. Indeed, Donna Haraway argues that the “radical constructivism” of the postmodernists falls into

the same traps of totalization that the mythos of disembodied science does through the “equality” of positioning” (Haraway 7, 11). However, to best understand and characterize the rhetorical operations and construction of the vaccine-hesitant epistemic sphere, this equality of positioning is necessary— thus, I will adopt the anthropologist’s idea of cultural relativism, in which actions made within any culture (or in this case, epistemological sphere) are best understood within their own cultural context. Baier’s work provides theoretical support for adopting this position in relation to his differends: He highlights Lyotard’s work *The Differend*, in which differend’s are “a case of conflict between (at least) two parties, that cannot be equitably resolved for lack of a rule of judgment applicable to both arguments” (“The Differend” xi). This conflict stems from a communicative failure because parties do not share metanarratives: indeed, “[e]ach genre of discourse would be like an island” (“The Differend” 130) (Baier 103). While Lyotard supposes that communication can be possible between these discursive islands, Baier finds his epistemic spheres to be “incommensurable” because the fundamental concepts of “truth or factfulness” are not shared (103). Applied to our current analyte of expertise and science, the alternate vaccine experts and the traditional vaccine experts become two epistemic ships, their paths charted so they do not intersect, sailing past each other in the dark of night.

A Case Study: Drawing Out Fragmented Expert Epistemologies

To understand how these alternative epistemic spheres arise, and attempt to allow for communication between them, I aim to examine the internal logic of the vaccine-hesitant epistemic sphere. Drawing from Anderson’s observations that in scientific knowledge, laypersons must often rely on second order judgements of expertise instead and the facts themselves, I focus my attention on the construction of expertise within these spheres. This

demands a site where biomedical authority is actively negotiated. COVID Vaccine Hesitant Discourse on The Joe Rogan Experience podcast exemplifies these needs.

Why COVID Vaccine Hesitancy?

The World Health Organization's SAGE Working Group on Vaccine Hesitancy defines vaccine hesitancy as a "delay in acceptance or refusal of vaccination despite availability of vaccination services" (MacDonald)². Vaccine hesitancy is a dynamic, evolving issue, which has existed in the West at least since the introduction of Edward Jenner's smallpox vaccine in the late 18th century (Spier), but it has taken on a new magnitude and ferocity in the wake of the COVID-19 pandemic (Ortiz-Prado et al.). With urgent ramifications for public health, vaccine hesitancy remains an urgent issue to address. It has been conceptualized in a number of forms—for example, the deficit model (Kitta and Goldberg)-- a model which posits that scientific information flows from the biomedical experts to an uninformed public, who are passively waiting to be filled with the knowledge they lack (Grant 1).

Despite the widespread use of the deficit model (Richardson and Segal), scholars have shown that vaccine hesitance stems from many other factors which are often quite rational. For example, Rhetorician Heidi Lawrence, in her book *Vaccine Rhetorics* (2020), draws out the limitations of conceptualizing of vaccine hesitancy as "simple deficit, a product of subscribing or not subscribing to the hegemony of science, or of 'delusional thinking,'" we lose sight of the "multiplicities of exigencies" (Lawrence 18). By these exigencies, she refers to the ways in

² I will note that the SAGE group makes a distinction between the terms vaccine hesitancy and outright vaccine refusal, but vaccine hesitancy introduces a broader scope of concerns and potential outcomes (ie, vaccinated but worried, worried and not vaccinated, considering vaccination with misgivings...). This broader scope makes the concept of vaccine hesitancy more applicable when it comes to conceptualizing around broad swaths of the population, not just the section of the population that outright refuses all vaccines.

which “objects and matter operate productively and with agency in the human world” (13), pointing to the way in which social and material realities influence thinking around vaccines. Exigencies such as vaccine injury, a phenomenon which we must remember does occur despite the widespread effectiveness and safety of vaccines, and the exigency of uncertainty, especially in relation to her interviews of young, healthy college students who are uncertain of the benefit the flu vaccine provides, are revealing. The latter example, viewed in the light of ever-cycling annual flu vaccine efficacy, helps us understand that these are not “deficits in understanding,” and “not as illogical, incoherent, or ignorant of science or risk statistics as they might seem at first glance” (116). The deficit model, positing a failure in logic and knowledge, is too simplistic.

Lawrence’s conceptual move away from the deficit model demonstrates that vaccine-hesitancy is not a failure in rational scientific understanding. Moreover, it allows us to understand how new, alternative-forms of knowledge are invoked in the vaccine-hesitance phenomenon. In her book *Calling the Shots: Why Parents Reject Vaccines* (2016), sociologist Jennifer Reich interviews parents who choose to delay or refuse vaccines, and finds that vaccine-hesitant parents can both be aware of scientific data around vaccine safety for the general populace, but do not necessarily view that data as the only consideration necessary for their particular child. They believe they “most intimately know their own children, understand their unique health statuses and vulnerabilities, and believe that they can most accurately estimate risk” (70). This demonstrates how biomedical knowledge and another form of claimed knowledge, knowledge of one’s own child, coalesce to create the vaccine-hesitancy. Similarly, sociologists Claire Decoteau and Paige Sweet examine the “accumulations of distrust” due to the legacies of medical racism, which make vaccine hesitancy (and broader biomedical rejection) a rational reaction (Decoteau and Sweet). This reflects a broader mistrust in institutions, and

demonstrates how a rational distrust of biomedicine can stem from the knowledge of past harms. In this way, vaccine hesitancy is not simply a rejection of mainstream knowledge but a construction of new forms of knowledge, placing the vaccine-hesitant phenomenon squarely within the fragmentation of epistemic spheres.

Why Joe Rogan? Why Podcasts?

Not only is social media a site of proliferating vaccination discourse (Skafle et al.), but the platforms themselves contribute to epistemic fragmentation. Historian Nancy Tomes writes about the “infodemic,”-- the surge of misinformation which spreads faster than COVID itself-- and how it can lead to differing conclusions amongst different viewers: “The line between information and infomercial can be hard to discern. The definition of good versus bad facts depends on the eye of the beholder. What is indisputably true to one group is seen as rank falsehood by another” (Tomes E1312). This conceptualization of relative truth points to the necessity of utilizing the postmodern epistemic sphere as a framework for analyzing social media discourse. Moreover, social media algorithms show different information to different people-- the presentation of different information fuels epistemic fragmentation. As Social Scientists Di Martino et al. write, social media leads to the “formation of ideologically homogeneous clusters of users, called echo chambers” because of how recommendation algorithms and user preferences interact; They call this the “ideological fragmentation” of social media (Di Martino et al. 1). This ideological fragmentation, because of its reliance on physically showing individuals different information due to the algorithms, points to a broader epistemic fragmentation; Indeed, as philosopher Matthew Crippen concurs, there is a “selective receptivity and blindness to information in the context of social media” which leads to “epistemic fragmentation” (Crippen

1). Social media, then, becomes an ideal place to explore how the presentation of different information contributes to epistemic fragmentation.

Moreover, examining these social media fragmentations with the methods of rhetorical analysis is particularly salient to understanding the epistemologies that emerge. As Communication Studies scholar Amy Koerber writes in *From hysteria to hormones: A rhetorical history* (2018), rhetorical analysis of media frames allows for the examination of:

the ways in which society, culture, and science interact to produce a medical belief that is convincing at a distinct moment in time, and at a distinct geographic location, because it resonates with its intended audience at that place and time, for reasons that can include historical and social realities just as much as scientific ones (Koerber 6).

Media exposes the beliefs that are persuasive to the audience, and thus, rhetorical analysis allows one to dive into what those beliefs are. Similarly, Lawrence writes in *Vaccine Rhetorics* under her aforementioned exigency model, “rhetoricians can produce the nuanced methods of research, engagement, and intervention needed to reclaim persuasion and discourse within such issues” (Lawrence, 18). This demonstrates the potent role of rhetorical analysis to not only draw out the existence of these alternative epistemic spheres and therefore the alternative-experts, but to draw out the ways in which the discourse can be changed and shaped in the future. This scholarship is crucially important for mainstream-experts to internalize, as understanding this construction and orienting science communication efforts around it is salient in cementing their expert status.

To examine expertise, specifically, in the context of vaccine and social media epistemic fragmentation, I will use the Joe Rogan experience as a case study. This is partially because his podcast is enormously popular and culturally impactful. Indeed, his podcast is one of the most viewed in the world, ranking #1 on Spotify and with 20 million Subscribers on youtube (“Joe Rogan Hits New Record as Spotify Reveals Most Popular Podcasts”; “PowerfulJRE’s YouTube

Statistics”). Moreover, he has a cultural impact: communications scholar Eli Kuhn examined the role he and other influences played in shaping the results of the 2024 US presidential elections (Kuhn). This popularity and impact ensures that the medical beliefs espoused there, no matter how erroneous, resonate with a significant swath of the US population. Moreover, Rogan’s podcast is a prime site to investigate epistemic fragmentation. As Science Communication scholar Dominik Stecula, notes, the Joe Rogan Experience operates as an epistemic community shaped by what scholars term “epistemic polarization”—that is, “a divergence in what different groups recognize as valid knowledge and trustworthy information sources” (8). In this way, Rogan’s platform becomes a site of active knowledge contestation, forming its own epistemic community, under Baier and Lyotard’s terminology. Thus, the Joe Rogan Experience is not only a popular podcast with a wide reach, but also an active site of knowledge contestation that is ripe to be further analyzed.

Finally, the Joe Rogan also supplies an ideal forum for specifically studying expertise under epistemic fragmentation because each of his podcasts are formatted the same—Rogan, a layperson, invites a prominent figure on the show who he chats with and asks questions of, over the course of two to three hours. These figures are upheld as credible, authoritative experts in their field. In late 2021, five guests with putative biomedical expertise appeared on the show to talk about COVID and COVID vaccines. These guests were Dr. Kory, Dr. Weinstein, Dr. Gupta, Dr. McCullough, and Dr. Malone.³ Two of the figures, Dr. Kory and Dr. Malone appear together in the same episode. All figures except for Dr. Gupta espouse sentiments which are skeptical, hesitant, resistant, or otherwise against the dominant discourse about vaccines, and will

³ The guests appear on the podcast in this order:

1. Dr. Kory and Weinstein appeared on June 22nd, 2021.
2. Next was Dr. Gupta: October 13, 2021
3. Subsequently, Dr. McCullough: December 13, 2021.
4. Finally, Dr. Malone: December 31st, 2021.

henceforth be referred to as the “alternative-experts.” Dr. Gupta’s traditional views on vaccines represent the mainstream scientific discourse around vaccination, and therefore he will be used as a comparison point and referred to as a “mainstream-expert.” These figures allow for the exploration of the construction of expertise in a discrete moment in time, on a discrete show.

Not only does Rogan’s high viewership indicate alignment if not outright agreement by many with these alternative-experts (one is simply unlikely to sit through a 2-3 hour podcast they disagree with)⁴, but the large reach and the real world influence of the figures who appear on his podcast and espouse antivaccine sentiments. For example, Dr. Malone was subsequently appointed to the Advisory Committee on Immunization Practices by RFK Jr., and was recently invited back on the show. Dr. Bret Weinstein has his own popular podcast, the Dark Horse Podcast, and has appeared on Rogan’s show multiple times, indicating the viewership was high enough on these specific episodes that Rogan thought it would be salient to bring him back.

I will focus on characterizing the rhetorical appeal of these emergent, expert epistemologies. Analyzing these episodes, I find that the alternative-experts generate credibility with lay audiences by reproducing and recasting the cultural imaginaries of insider biomedical expertise. In this way, expertise is rhetorical, and the potency of the persuasive appeal is what imbues the biomedical expert with their authority.

Chapter 1, “The Inside Man,” examines how the alternative-experts reproduce the cultural imaginaries of insider expert status. I use the term cultural imaginaries to denote the modes of understanding of biomedical expertise that are widely recognized in the west, and thus the reproduction of these imaginaries in the alternative-experts allows them to be positioned as biomedical insiders. I draw on the symbolic capital of biomedical expertise as theorized by Pierre Bourdieu to understand the potency of these cultural imaginaries. Specifically, his

⁴ Unless they are writing an honors thesis!

concepts of symbolic institutionalized capital (degrees and academic credentials) and embodied capital (the internalized culture of scientific and clinical practice) provide a framework to understand the appeal of the alternative experts. I also examine the comparative failure of Dr. Sanjay Gupta to reproduce this capital in the context of Rogan's podcast, demonstrating that insider status is ultimately rhetorical— even mainstream-experts— rely on the rhetoric of expertise to be recognized as an expert by laypersons. Secondly, I demonstrate how the alternative experts reproduce the culture of biomedical expertise and situate themselves within it. Consensus and communal validation are important cultural imaginaries of what science is and what biomedical expertise is situated within, so by reproducing this culture and situating themselves within it, alongside the first strategy, the alternative-expert maximizes their inside status.

Chapter 2, "The (Faux) Outsider," examines how outsider appeals are actually better understood as a transcendent, balanced distance between insider and outside status, and as a recasting of the cultural imaginaries of insider status with the populist and democratic ideals. This creates a calibrated dual positioning which enhances their expert status. Whereas both the cultural imaginaries of science and the populist and democratic ideals are held in the public consciousness of the American populus, the populist and democratic recasting has particular rhetorical appeal to Joe Rogan's viewership. This makes this rhetorical strategy particularly appealing for his crowd. I argue that I identify three key rhetorical moves which aid in this complex rhetoric of positionality: the construction of a transcendent distance from mainstream institutions that maintains credibility while claiming dissent; the delegitimization of the mainstream as a corrupt, profit-driven elite antagonistic to the American people; and the construction of an alternative epistemology that elevates populist-friendly forms of evidence such as anecdote, meta-analysis, common sense, and democratized knowledge production as a

new, populist epistemology. This demonstrates that alternative expertise is not a symptom of anti-intellectualism or scientific ignorance, but a sophisticated rhetorical move which situates insider and outsider expertise together.

1. The Inside Man: Reproducing the Cultural Imaginaries of Biomedical Expertise

Stereotypically, vaccine hesitant figures are painted as uneducated and unacademic. Yet my analysis finds that alternative experts are actually quite well educated, and have incredible experience and credentials. I find that they reproduce the academic capital and culture of biomedical expertise. The specific forms of symbolic capital and culture that they invoke are part of the broader cultural imaginaries (the stories that shape how society sees reality) that help us determine what expertise *should* be. By reproducing these cultural imaginaries, alternative experts gain their expert status. Appearing to be an *insider* of biomedicine, by claiming expert status, increases their credibility. This also reveals something vital about expertise: it is a rhetorical position that can be invoked by reproducing these cultural imaginaries.

When it comes to scientific matters, there is nobody better to turn to than an expert. So to be an expert, or at least seem to be one, is vital to claiming expertise. This insider status, where one is able to claim expertise and appear authoritative, naturally lends itself to creating credibility. The scholarly community recognizes how insider expert status has previously been weaponized to gain credibility and sway laypersons. The most famous example of this is in *Historians of Science* Naomi Oreskes and Erik Conway's study of Climate Change discourse in *Merchants of Doubt*. These authors write about how physicists, astrophysicists, and oceanographers who worked on the atomic bomb and studied space exploration became the prominent influential dissenters regarding climate change science. While "these men did almost no original scientific research on any of the issues on which they weighed in...." they were nevertheless able to use "their scientific credentials to present themselves as authorities" (Oreskes and Conway 8). These figures gained expert status and credibility by reproducing

insider expert status. I argue a similar dynamic occurs in the vaccine-hesitant community. Specifically, the vaccine-hesitant experts on the Joe Rogan Experience Podcast, Dr. Robert Malone, Dr. Peter McCullough, Dr. Pierre Kory, and Dr. Bret Weinstein, utilize scientific credentials such as their academic degrees, as well as their experience and roles as physicians and scientists, to construct insider status and thus gain expertise.

Academic credentials are a broad category, and I argue that two distinct rhetorical strategies work in tandem to situate these figures as biomedical insiders. The first rhetorical strategy is reproducing the symbolic capital which marks individual biomedical experts, such as academic qualifications and the embodiment of the culture of medicine. Relying on Pierre Bourdieu's conceptualization of the numerous spheres of symbolic capital, I highlight how institutional and embodied symbolic expertise allows for the alternative-expert to internalize the role of the scientist. The second strategy is reproducing the broader culture of science itself as the sphere in which the alternative-experts are situated, mostly by relying on the culture of consensus and communal validation of the putative expert. This internalization is salient, but an even simpler claim, that I am in the community and therefore I am an insider, is also just as potent— especially considering the privileged role of consensus in the biomedical community. Therefore, the second half of this chapter will explore how the alternative experts position themselves within the broader culture of biomedicine. Specifically, they reproduce ideas of the value of science as a consensus based, communally validated knowledge making practice or culture, allowing them to rhetorically become insiders. In this way, they occupy a rhetorical position of expertise.

However, many would argue that they do not represent the true consensus position, or fit within the broader, mainstream culture of biomedicine by aligning with the vast majority of

scientists. This point is moot, because ultimately it is the impact that matters, and the impact is that these figures have gained a privileged position in swaying layperson opinion. The mainstream-scientific community must internalize this finding if they wish to reclaim expert status and authority amongst the American public.

Symbolic Capital of Academic Qualifications & Credentials

Biomedical science is recognized by laypersons through a variety of forms of academic-based symbolic cultural capital. The reproduction of this capital by the alternative-experts allows them to be perceived with expertise. Symbolic cultural capital is a potent force which, as I will argue, allows laypersons to recognize biomedical expertise. Sociologist Pierre Bourdieu famously defined capital as something, that when “exclusive[ly]” held, has the capacity to “appropriate social energy” (Bourdieu 15) Money typifies capital, as it is deployed by those who have it to purchase other assets, and its power comes from the fact that some possess it while others do not. However, Bourdieu also postulates the existence of non-material forms of capital which confer a variety of societal purposes: There is cultural capital, which can function as institutionalized cultural capital, in the form of “academic qualifications,” and in the form of embodied cultural capital, which are “long-lasting dispositions of the mind and body” (17).⁵⁶ Cultural capital is symbolic because of its invisibility relative to material capital, such as money. This invisibility also makes it more potent: “Because the social conditions of [cultural capital’s] transmission and acquisition are more disguised than those of economic capital, it is predisposed to function as symbolic capital, i.e., to be unrecognized as capital and recognized as legitimate competence” (18). This demonstrates the potency of these forms of capital as an invisible, legitimizing force.

⁵ He also outlines material, “objectified” cultural capital, which are “cultural goods” such as pictures, books, dictionaries...” which is not relevant for the purposes of this argument but does exist (17).

⁶ To better understand the potency of the various forms of capital, consider the example given by Science and Technology studies scholars Mathieu Albert and Daniel Lee Kleinman in the article “Bringing Pierre Bourdieu to Science and Technology Studies:

“In the scientific field, economic, social and cultural capital may play a significant role in scientists’ hierarchization: economic resource may enhance one’s academic prestige by allowing one to hire more graduate students...degrees from prestigious institutions can open doors, but equally having internalized as one’s own habitus—one’s way of being in the world—the legitimate culture of the scientific field (knowing what to say, what book to have read, who to cite, etc.) can offer pivotal advantages too.” (Albert and Kleinman 268)

To further explore the usage of this capital in establishing expertise, in the first part of this section, I dive into the usage of the degrees as an institutionalized form of scientific capital (1.1.1). I later explore the reproduction of the embodiment of biomedicine as capital, which demonstrates an internalization of the culture of academic biomedicine on the part of the alternative-expert (1.1.2). Whether intentionally or not, these individuals reproduce the markers denoting traditional biomedical academic capital, thereby increasing the perception of their expertise. Finally, I dive into the ways in which the mainstream-expert on the podcast, Dr. Sanjay Gupta, fails to reproduce the symbolic capital of medicine, and thus does not succeed in establishing his own expertise amongst Rogan's audience (1.1.3).

Degrees as Institutional Capital

Academic, degree-based credentials are a form of symbolic capital used to denote biomedical expertise. Despite the common perception that those who are involved in vaccine-hesitant discourse are uneducated, all the alternative-expert guests on the podcast invoke terminal, impressive educational credentials through an MD, Ph.D., or both: Dr. Kory and Dr. McCullough both have MD's, Dr. Weinstein has a Ph.D., and Dr. Malone has both an MD and a Ph.D. By invoking these degrees, the alternative-experts reproduce a particular form of symbolic capital that Bourdieu calls institutional capital. Bourdieu defines academic qualifications, also known as degrees, as a specific form of symbolic cultural capital known as institutional capital (17), which is an especially potent form of capital because it bestows recognition. Degrees function as "a certificate of cultural competence which confers on its holder a conventional, constant, legally guaranteed value with respect to culture" which is "institut[ed] by "collective magic" (20). This collective magic stems from the associations we have with academia as "one has only to think of the concours (competitive recruitment examination)" to begin to envision it

as an exclusive differentiator, and thereby something with capital (20). With degrees “one sees clearly the performative magic of the power of instituting, the power to show forth and secure belief or, in a word, to impose recognition” (20). Therefore, there is great power in possessing degrees.

Aside from their exclusivity, the US culture of professionalization grants this form of symbolic capital its potency as a marker of authoritative biomedical expertise. Professions are a sociological concept distinct from a job or occupation (Monteiro, 49). According to Chemist Dr. Kovac, a profession’s members “lay claim to specialized knowledge and skills,” because of the “standards of education and training, often culminating in a formal certification[s],” This demonstrates how education functions as an indicator of scientific skills and knowledge, and plays a role in establishing scientific expertise (210). Across his work, Kovac argues that sciences should be recognized as professionalized because of their conformity to this dynamic, pointing out the unifying education of a Ph.D. (210). This concept of professionalization similarly applies to physician expertise and authority. In *The Social Transformation of American Medicine*, Paul Starr argues that physicians “claim authority, not as individuals, but as members of a community that has objectively validated their competence,” which guarantees that individual practitioners have a high level of skills and competence, which lend themselves to increasing expert authority (12). These desirable qualities, which indicate expertise, are “institutionalized in a system of standardized education and licensing” (Starr 19), demonstrating how education (and the symbolic prefixes associated with them- ex Ph.D., M.D.) becomes an important proxy by which status is procured. Because of these cultures of professionalism, education functions powerfully as symbolic capital denoting authoritative competence and skill, lending itself to the establishment of expertise in biomedical domains.

The symbolic capital of education is further conceptualized as meritocratic and thus democratic, heightening the appeal to expertise and authority in a country that prides itself on (supposedly) being democratic. Starr demonstrates the value of education in our context by juxtaposing the basis of 19th-century emerging physician professionalism in the US with the basis of English physician professionalism: “The esoteric learning, knowledge of Latin, and high culture and status of traditional English physicians were more compelling grounds for belief in a hierarchically ordered society than in a democratic one,” so professionalism had to be “reconstructed around the claim to technical competence,” through “standardized training and evaluation” (17). Implicit in his claim is that technical competence, gained through education is meritocratic since meritocracy and democracy both supplant hereditary privileges with a system that prides itself on success through hard-work and talent, where the values of education and democracy are aligned. Because meritocratic values of the symbolic capital of education have a potent role in the democratic consciousness of the US, it cements the appeal of expertise via education in our context.

The impact of education as symbolic capital on conferring expertise has been previously demonstrated in vaccine hesitant communities. Statisticians Harris et al confirms the prevalence of using educational degrees as a stand in for expertise in vaccine hesitant discourse in the United States. Using statistical methods, they find that the inclusion of academic prefixes (Dr., professor) or suffixes (MD, RN...) and other biographical descriptions which imply experience in a biomedical field are influential in procuring engagement (operationalized as likes, retweets, etc) on Twitter/ X in tweets which engage in vaccine-hesitant discourse. This demonstrates that academic degrees are influential in supporting expert status amongst laypersons.

Since education functions as symbolic capital conferring expertise, the prominent use of educational degrees by the alternative-experts on Rogan's podcast increases their expert status. In a professionalized context, these degrees are used as heuristics through which the audience can ascribe expertise. Importantly, the symbolic capital of degrees cannot be taken away, further cementing the expert authority of these alternative-experts even when their retractors point out flaws in the information presented. While Rivera, the physician who organized the 270 mainstream-experts, notes that Rogan has a "patina of academic credibility," suggesting the invalidity of his academic credentials to discredit him, the existence of his credentials weakens Rivera's claim. As Bourdieu writes, academic qualifications, as a form of institutional symbolic capital, are permanent: they "institute[s] an essential difference between the officially recognized, guaranteed competence and simple cultural capital, which is constantly required to prove itself" (20).

The appeal of the institutionalized capital of academic qualifications to laypersons becomes evident in Joe Rogan's remarks to Dr. McCullough. As the host of the show, Joe Rogan is not positing himself to be the putative expert, but instead is engaged in a conversation with the alternative-experts. In this way, he functions as a layperson standing in to represent the audience, allowing us to ascertain the reactions his anonymous viewers have that underlie their support for the alternative-experts. Rogan's opening remark to Dr. McCullough, made in media-res, is that some of his medical videos have been taken down from YouTube: "I found that very odd— that a doctor talking about a medical disease would have videos taken down-an actual expert!- would be either testifying or discussing treatments and talking about a disease and have those videos taken down off of YouTube" (Rogan, "McCullough," 00:25-00:48). Rogan's impassioned injunction of "an actual expert!" relies on the fact that Dr. McCullough is a medical "doctor

talking about a medical disease.” This demonstrates how in a professionalized context, physician competence is perceived as standardized, discouraging critical inquiry into their advice and messaging. It speaks to the standardized competence assumed of doctors which Starr discusses, because it fails to acknowledge that Dr. McCullough, as a physician with an MD degree, could be incorrect about a medical disease. Additionally, Rogan’s interaction with Dr. McCullough glosses over the specific content that was removed. This demonstrates a willingness to focus on educational credentials alone even if it means refraining from exploring the granular details of the content in the video and the circumstances for why they were removed. This is a prime example of exactly how this capital functions as a mental-short cut, or heuristic, which supersedes any need to validate or ensure accuracy, thereby increasing the alternative expert’s status as experts.

Though the mere possession of academic credentials is sufficient to invoke expert status—as the above interaction between Rogan and Dr. McCullough demonstrates, the doctor and disease match is nevertheless appealing to Rogan. Expert status is improved when the practitioner’s education specifically maps onto the field in which they claim expertise. In other words, though doctors all know something about a variety of medical conditions, ideally you don’t want a psychiatrist performing your knee replacement, and you don’t want an orthopedic surgeon telling you why you’re sad. Indeed, human sciences researchers Brand et al’s study on prestige-based learning demonstrates how people do attend to domain specific cues and attempt to choose experts based on a perceived match between subject and domain of expertise (Brand et al.),⁷ demonstrating the desirability of this match in content and expertise. Consequently, the

⁷ This study also finds that they will also choose to trust someone with prestigious credentials in an unrelated field to the topic at hand over a random individual, showing the generalizable power of a degree is still substantial when compared to someone who doesn’t have a specific cue (Brand et al.)

specificity between symbolic educational credentials and topic discussed, may help explain why these Joe Rogan episodes were so well received and the perception of the alternative-expert status was increased.

This specificity gives the institutional degree more weight as capital– it is not only a science/medical degree, but a degree which specifically applies to COVID and COVID-vaccines. Clinical experience can be used to map specificity between expert fields and expertise. Two of the physicians actively practice clinical medicine– Dr. Kory introduces himself as a “lung and ICU specialist” (“Kory & Weinstein” [00:00:39-00:00:43]), and Dr. McCullough presents himself as a “internist and cardiologist” who is “also trained in epidemiology” (“McCullough” [00:00:50 – 00:00:56]). Pulmology, intensive care and epidemiology have clear relevance to a pandemic with a significant respiratory component which infamously led to surges in admissions in ICUs across the country. Ostensibly less related is Cardiology, however, in 2021 (before the podcast episodes discussed were made) The Ottawa Institute would release a preprint of a paper showing a link between the COVID vaccine and a type of heart disease called myocarditis in children (Kafil et al.). Though this preprint was eventually retracted, vaccine-induced Myocarditis would become a central anxiety in the vaccine-hesitant community, and thus cardiology training appeared more relevant. In this way, the alternative experts map specificity between their fields and expertise. The invocation of specific experience, both clinical and research based, increases the potency of their degrees as institutional capital.

Aside from clinical experience, research experience can provide specificity between degree and expertise. Dr. Malone and Dr. Weinstein, the MD-PhD and PhD (respectively), present themselves through the lens of research science. Dr. Malone applies his MD-Ph.D. to working on mRNA vaccine technology and responding to disease outbreaks, creating an

instantly recognizable connection between his training and mRNA COVID vaccines/ pandemic management (“Malone” [00:01:30 - 00:08:55]). As opposed to solely relying on the general educational credential of an MD or PhD, the specificity improves his perceived expertise. Similarly demonstrating his specific expert relevance to COVID, Dr. Weinstein introduces himself as an “evolutionary biologist,” a field which arguably is not particularly related to COVID from a health perspective (“Weinstein & Kory” [00:02:11-00:02:14]). Yet it is not actual applicability, but rather perceived applicability that matters. Dr. Weinstein ties his field to COVID by discussing the evolutionary basis of viruses, epidemiology, and the immune system, concluding that:

The importance of evolution here is... all of the things that we're talking about with COVID are evolutionary. Obviously, the virus is evolved. Epidemiology is an evolutionary, uh, process. The immune system is both a product of evolution and it evolves in real-time when you have an infection. So evolution is the kind of good generalist toolkit to apply to something like COVID (“Weinstein & Kory” [00:02:15 - 00:02:39]).

This sheer repetition of the word “evolutionary” frames many facets of COVID biology and science under Dr. Weinstein’s field of putative expertise. In actually, his research is about evolutionary trade-offs related to aging and tumors (Weinstein and Ciszek) His rhetoric is akin to saying, I study the chemistry of organic life forms, and thus I am an expert on all of life, when in reality most of your day to day in a lab is probably studying carbon chains. Despite this, the rhetorical appeal of this specificity is persuasive.⁸

However, while anybody can have experience, proving their experience is exceptional or they are exceptional examples of individuals in their fields further increases the

⁸ The scientists also rely on prestige to bolster their academic qualifications. Indeed, Dr. Malone liberally invokes names such as Harvard, Northwestern, and the Salk Institute when discussing his training (“Malone” [00:01:30 - 00:08:55]) . Moreover, Dr. Weinstein is described as a “visiting fellow at Princeton” in the description of the podcast (“Weinstein & Kory”) However, this fellowship appears to be in the James Madison Program in American Ideals and Institutions, not a biological fellowship (*Culture vs. Consciousness*). Despite the fact that this pertinent information about the fellowship is obscured and never mentioned in the podcast, the prestige of the institutional academic qualifications is still improved.

alternative-experts' rhetorical appeal to laypersons. This is made manifest in the extremely impressive research accomplishments. An important symbol of biomedicine is the scientific study or paper. As a discrete unit of research which is widely recognized by the public, they become potent symbols of the broader research endeavor. This symbol, representing the broader biomedical mission, functions as a material cue of insider status. This is demonstrated in Dr. McCullough's assertion that he is "the most published person in [his] field, in history, with over 650 publications in the National Library of Medicine," ("McCullough" [00:01:23 – 00:01:31]) and "51 publications on COVID-19" ("McCullough" [2:35:05 – 02:35:16]). Indeed, google scholar shows he is an author on papers with thousands of citations, the largest being 6022 citations as of April 2026 (Google Scholar, "Peter A. McCullough"). This, frankly absurd, number of papers invokes the numerosity heuristic, more colloquially referred to as "bigger is better," to imply his quality as a researcher. This point is reinforced in Dr. Malone's podcast episode with Rogan, where Dr. Malone states that Dr. McCullough is "is one of the most highly published authors in the world, um, he's an exceptional researcher " which cements Dr. McCullough's perceived research status as a singular contributor to the field ("Malone, [01:14:50 – 01:15:10]). Dr. Malone similarly asserts his own biomedical expertise by invoking the plentiful trials he ran: he asserts that he "[ran] over 100 clinical trials mostly in the vaccine space" where he "won literally billions of dollars in federal grants and contracts" ("Malone" [00:01:30 - 00:08:55]). Quantity of studies behaves as a heuristic for quality, and these exceptional numbers for both researchers create a perception of exceptional research work. Moreover, Dr. Malone uses the term "won" funding, which implies success in a competitive funding landscape. A scientist is unlikely to procure billions of dollars of funding with poorly designed or marginally applicable studies, which increases the perception that his research is exceptional. The relevance of his

work to vaccines furthers his status. It is important to consider that these figures are in conversation with each other. Dr. Malone, by asserting Dr. McCullough's publication support, lends his own credibility to Dr. McCullough. As this chapter will later discuss, support from other researchers and credible figures ultimately increases perceptions of expertise.

Exceptional research credentials are also demonstrated by invoking prestigious journals for the papers/ trials they associate their work with. Discussing his COVID drug treatment protocol, Dr. McCullough writes, "We submitted our paper July 1st to the American Journal of Medicine, which is one of the highly ranked journals in medicine" ("McCullough" [00:07:13 – 00:09:17]). While there is a distinction between submitting something for consideration and being accepted by a journal, this distinction gets blurred by such a claim. Thus, the work gains the benefit of association with a "highly ranked" journal, even if the journal later rejects it– as the layperson will be none the wiser.

Similarly, Dr. Malone's allusions to "the National Library of Medicine," allows him to bolster his research reputation by invoking a seemingly important source. Although the NLM is technically a database, not a specific journal, the words "national" and "library" imply scholarly credibility, and publication within this source implies publication quality in the same way that scholarly journals would convey credibility through publication. The first Google result for the string "National Library of Medicine" is the NLM page, which is plastered with the logos and affiliations with the National Institute of Health (NIH), and the hyperlink includes a .gov domain, further implying credibility of the resources found within it through affiliation with the US Government and the NIH. However, since NLM is a large database, it is essentially redundant for Dr. McCullough to include it– as long as his work is published and biomedical, it will likely appear in NLM. Nevertheless, the lack of substance in the statement is simply glossed over. In

this way, Dr. Malone and Dr. McCullough not only fulfill the embodiment of a scientist through their research activities, but the seemingly incredible nature of the accomplishments afford the alternative-experts an exceptional embodiment, furthering their status as experts.

However, the astute reader will have noted the many technicalities and qualifications which make the seeming impressiveness of the journals less powerful— this demonstrates how these invocations are ultimately rhetorical tools, and can be misleading when it comes to identifying ontological expertise.

Yet, simply possessing the academic qualifications of degrees is only the first step. The next section will explore how the embodiment of biomedical expertise functions as another form of capital which gives the alternative-experts insider status.

Embodied Academic Qualifications

While the degree itself is institutionalized capital, the accumulated, lived experience and skills of scientists and physicians represent what Bourdieu calls embodied capital— cultural capital which is “linked to the body,” and is “external wealth converted into an integral part of the person” through a process of “embodiment” and “incorporation” which “implies a labor of inculcation and assimilation” and “costs time, time which must be invested personally by the investor” (17). These requirements can be understood as the time and effort put into not just earning the degree, but actually integrating what it means to be a scientist or a physician into the self, *becoming* this role and fully embodying it. Bourdieu associates embodied capital with what he calls the “habitus” (17). To better understand the habitus and embodied capital in relation to the biomedical expert, consider the example given by Science and Technology studies scholars Mathieu Albert and Daniel Lee Kleinman in the article “Bringing Pierre Bourdieu to Science and Technology Studies: “In the scientific field... having internalized as one’s own habitus—one’s

way of being in the world—the legitimate culture of the scientific field (knowing what to say, what book to have read, who to cite, etc.) can offer pivotal advantages too.” (Albert and Kleinman 268) This demonstrates the role of the habitus— the internalization of the culture of science— as a way of being for scientists which functions as embodied capital and thus generates rewards for the actors who utilize it.

While Albert and Kleinman’s argument concerns the capital function of embodied habitus within the internal workings of the scientific community, I argue that this cultural capital of an embodied biomedical way of being creates the perception of expertise. As Starr writes, "scientific knowledge has held a privileged status in the hierarchy of belief," demonstrating the importance of the scientific way of knowing for the public (5-6). Thus, the ideals of science which produce this privileged status, when embodied by an actor, lend that actor credibility and the perception of expertise. I argue that certain cues of the ways of being which are widely recognizable and visible to laypersons allow for this embodied capital to be reaped, thus, the alternative-experts are imbued with expertise.

The internalization of the cultural norms and values of science also serve as capital through which expertise is increased. Crucially, these ideals are not confined to the internal culture of science, they are publicly recognizable. Ranalli, in "Climate Science, Character, and the 'Hard-Won' Consensus," demonstrates that Sociologist Robert Merton's scientific norms function as widely shared heuristics through which both scientists and laypeople evaluate scientific credibility, finding that participants on both sides of the climate science debate intuitively invoke Mertonian ideals, particularly disinterestedness and organized skepticism, to assess the character and trustworthiness of scientific figures (Ranalli, 188). The alternative-experts orient their argument among these publicly visible norms. As Ranalli

describes them, they represent the expectation that scientists are "professionally humble and disinterested...turn their skepticism inward and examine their biases...set aside ego and material interests" and "strive for objectivity and be willing to follow the evidence wherever it leads" (189).⁹ When a scientist is perceived as embodying this norm of disinterestedness, their credibility increases in the eyes of lay audiences.

Like the climate skeptics, the alternative-experts rhetorically invoke this ideal of disinterestedness, and it increases the embodiment of their biomedical expertise. For example, Dr. Weinstein, discussing his Dark Horse Podcast with his wife Heather Heying, which has recently been removed from youtube, states "Heather and I... what we are doing is we are showing our work and when we get something wrong, we are dedicated to going back and correcting it so that people who are trying to track our model of things get the update, right?" ("Weinstein & Kory" [00:29:45 - 00:30:11]). During this quote, Rogan periodically interjects with an affirmative, "yes", demonstrating the rhetorical appeal of Dr. Weinstein's claims on Rogan, who functions as a stand-in for the lay audience. Similarly, Dr. Weinstein later discusses the evolution of knowledge of how viruses evolve in the context of his disinterest in the outcome: "I remember going out of the house wearing cotton gloves, sacrificial gloves... I stopped doing that almost instantly as it became clear that actually, although many viruses do transmit from surfaces, this one doesn't" ("Weinstein & Kory" [00:31:17 - 00:31:37]) Both of these quotes demonstrate a willingness to pivot and discard previously held beliefs in the light of new data. As opposed to clinging to previously held beliefs, this willingness to follow the

⁹ The other norms, as described by Ranalli pulling from Merton, include: "universalism, communalism ("communism" in Merton's original formulation), disinterestedness, and organized skepticism. Universalism requires that contributions to scientific discourse be judged on their merit, without regard for the contributor's nationality, race, religion, sex, class, and so forth. Communalism makes imperative the free and open communication of methods and results. [D]isinterestedness requires practitioners to disclaim or forgo any egoistic or material stake in the outcome of their research. organized skepticism (sometimes simply called "skepticism") refers to the right and duty of the scientist (in Merton's formulation, the scientific community at large) to question any and all dogma" (Ranalli 186).

evidence exemplifies disinterest. Both in his podcast and in his own life, Dr. Weinstein fulfills the embodiment of a biomedical scientist through performing disinterestedness, thereby gaining the perception of expertise.

Aside from scientific norms, another recognizable cue indicating the internalization of biomedical culture is the utilization of scientific papers. Science and Technology Scholars Bruno Latour and Steven Woolgar demonstrate the important cultural role of the paper in their ethnographic study of a scientific lab, *Laboratory Life: The Construction Of Scientific Facts*. They conclude that writing and scientific papers are important because of the role they play in the construction of scientific knowledge: “skill is only a means to the end of publishing a paper ... The production of papers is acknowledged by participants as the main objective of their activity” (66). This is because papers act as a communication device, a “means of communicating ‘important findings’” and “facts” (71). In other words, since a scientist uses papers, then embodying the culture of science relies on using papers. Physicians’ culture similarly relies on papers, because, according to Starr, the profession of medicine “enjoys close bonds with modern science” (4). Consequently, reproducing this embodied capital allows one to be situated within the culture of biomedicine.

The alternative-experts, by liberally invoking papers, situate the way in which the culture of biomedicine is internalized within them, and thus successfully gain the expert status granted from this form of embodied capital. For example, Dr. Malone, discusses a “recent paper out of Hong Kong,” a “comprehensive analysis” of “myocarditis in boys hospitalized,” invoking the paper to support a claim he makes about vaccination causing higher rates of myocarditis and hospitalization than COVID itself. (“Malone” [01:22:54 – 01:24:00]). Dr. Malone invokes the same paper, citing a specific hospitalization rate of 1 in 2,700 which adds impact through

drawing on statistical data [01:23:08 – 01:23:22]. They claim, in contrast that the mainstream-experts claims that myocarditis is “mild” and thus not something to worry about are unsupported by “facts”, and only using “speculation”, because there haven’t been any long term studies ([01:23:22 – 01:24:00]). He also cites “over 140 studies that document that natural immunity is superior to vaccine-induced immunity,” counting on the sheer volume of literature on this topic to reinforce his claim. Dr. Malone references a Harvard study from Brigham and Women's nurses on spike protein circulation after vaccination to support his claims about the vaccine's systemic effects, relying on the prestige of these named institutions to further bolster the study. Dr. Kory similarly invokes a study conducted on the effects of “monoclonal antibodies” to prove their efficacy. The sheer magnitude of these studies illustrate how these alternative-experts are seemingly steeped in the culture of biomedicine, allowing them to profit from the embodied capital of an expert.

Moreover, it is not only that they invoke papers and other studies, but clearly describe the scientific mechanisms behind the claims that are being made. Invoking scientific evidence and logic both improves the embodiment of a biomedical expert, as a scientist/ physician is expected to base their claims on scientifically logical bases, but simultaneously, it creates transparency and accessibility that is appealing to the layperson. As chapter two will further explore, the common sense, accessible logic that underlies these claims increases their appeal.

Mainstream experts would object that many of these studies and this scientific logic was later challenged, problematized, or outright retracted, or that the studies were simply misused by the alternative experts. Nevertheless, the vast numbers of these papers drowns out the rhetorical counter-claims, as fact checking all of the studies and their uses is impossible for the casual viewer. Ultimately, they remain a salient rhetorical tool despite this rebuttal, demonstrating the

alternative-expert's internalization of the culture of biomedicine through their use of papers and scientific information, and how they benefit from the embodied capital they denote allowing them to be considered as experts.

Alongside the cultural imaginaries of scientific evidence, the culture of biomedicine also features a substantial, well recognized orientation towards actively healing, making invocations of healing a recognizable cue of a physician's internalized culture. As healing and maintaining the health of others is the *raison d'être* governing a physician's way of being, an orientation towards healing conveys the embodiment of a physician. Indeed, Starr recognizes the utility of this value and the orientation towards health as a basis of physician professional authority, arguing "[p]rofessional authority also presumes an orientation to specific, substantive values-in the case of medicine the value of health. Insofar as a practitioner violates those values... the exercise of authority is understood to be illegitimate" (Starr 12-13). The theory of embodied capital explains why the relationship between health and authority is "understood" by laypersons: this embodiment functions as a form of currency, with a potency lending authority and expertise.

Simply invoking their medical specialization, (ie, describing their physician roles as cardiologists, ICU doctors, etc, as discussed at the end of Section 1.1.1) demonstrates an orientation towards health through relevant clinical experience; However, anecdotes are an even more potent way of demonstrating this capital, as they can also help the viewer visualize the fulfilment of these embodied roles. Moreover, anecdotes allow for passionate, emotional expression, which improves the efficacy of rhetoric. This emotional rhetoric powerfully insists upon the alternative-experts fulfilment of physician embodiment. Powerfully demonstrating this health orientation and clinical experience, Dr. Kory states, "My first case of a patient who I

treated for long COVID, I mean, they literally were almost crying in joy because they had been sick for so long" (Weinstein & Kory" [02:08:24 - 02:08:34]). Beyond just simply stating he treats patients, the profound impact conveyed from healing someone who was "sick for so long" and their tearful response adds an emotional depth and profoundness which enhances the perception of his embodiment as a physician. Similarly, Dr. Kory's assertion that "There's nothing more satisfying than bringing them back" demonstrates a proud commitment to the role of healer, and alludes to the patient's close brushes with mortality, imbuing Dr. Kory with an authority that comes from demonstrating both a profound commitment to healing his patients and to his profession ([01:05:30 – 01:05:54]). Dr. McCullough similarly procures emotional anecdotes to convey his embodiment as a physician and biomedical expertise. For example, he states "once I turn my phone on from this interview, Joe, I am going to be loaded with cases that I'll hear, advise, try to help on, all the way home," painting himself as a passionate physician with a willingness to continue treating patients with COVID even off the clock ("McCullough" [02:07:38 - 02:07:45]). This rhetoric of care, healing and commitment allows the alternative-experts to possess the embodiment of a physician, increasing their expert status.

These examples demonstrate how the embodied capital of lived scientific and clinical practice are invoked by the alternative-experts. Taken together, the institutionalized capital of academic qualifications and the embodied capital of clinical and research practice operate as two forms of symbolic capital which alternative-experts reproduce to gain insider status. The referencing of the academic degree confers a permanent credential interpreted positively through the culture of professionalism, and the embodied performance of research and healing animates that credential. Utilization of these two rhetorical strategies demonstrate that the biomedical expertise has been genuinely incorporated into the self.

The efficacy of this dual invocation is demonstrated in Rogan's responses to Dr. Malone. As a non-scientist or physician layperson, Rogan functions as a proxy by which we can evaluate the effectiveness of the alternative-expert appeals. Echoing earlier remarks about censorship of Dr. McCullough's YouTube channel, Rogan says that Dr. Malone's Twitter ban is "very confusing," because of his extensive education and experience with vaccine research and outbreaks. Based on rhetoric, Rogan has concluded that Dr. Malone is "one of the most qualified people in the world to talk about vaccines" ("Malone" [10:03– 10:08]) This reproduction of the symbolic forms of capital in academic qualifications and embodiment capital thus reproduce the insider status of the biomedical expert by invoking the cultural imaginaries of expertise. This ultimately functions to create expert status in the mind of the layperson.

Finally, it is important to touch on the important role of gender and race in performing embodiment capital. As caucasian men, they reflect the stereotypical embodiment of scientists and physicians. Alongside resonating with Rogan's audience, which is predominantly white and male, they appeal to the broader cultural ideas of embodiment of people in STEM. This embodiment will be fairly obvious to the consumer of the podcast, as the Rogan podcasts are displayed on youtube and spotify in video form, and even those who opt to only listen will likely see the thumbnails of the podcast, which include the guest speaker. This form of embodiment capital, though less visible in direct quotes, subtly creates expert status.

Unsuccessful Invocation of Academic Credentials

Although the symbolic capital which confers expertise is based around traditional notions of science and scientific expertise, this does not mean that mainstream-experts can successfully reproduce it. Ultimately, this capital is rhetorical, and mainstream-experts need to be able to invoke it in order to construct their expertise for those who view Rogan's podcast. Dr. Sanjay

Gupta provides us with an example of a mainstream-expert who fails to invoke the academic and research-based rhetoric which confers academic status. I argue that this explains why he is rejected by Rogan's viewers and not considered by them to be an expert.

On the one hand, he has similar symbolic institutional credentials as the alternative-experts, which supports his mainstream-expert status: Despite not being explicitly asked by Joe Rogan to introduce himself, something which all the other guests are asked to do, Dr. Gupta mentions in the course of his opening conversation with Rogan (which starts in media-res) that he is "a practicing physician," demonstrating his educational credentials in the form of an M.D. ("Gupta" [00:08:21 – 00:09:02]). Additionally, later on in the episode Dr. Gupta states that he is a neurosurgeon- a role generally understood to be held by only the smartest and most competent physicians. Both revelations likely increased his status as an expert ([00:36:41 – 00:37:22]).

Regardless, he does not possess the strength of the embodied capital that the alternative-experts do, which weakens his expert status. He does not affiliate himself with COVID in a clinical or research capacity, as the alternative-experts do, so his expertise is less pronounced. His role as a healer is understated, as he actually says explicitly he does not take care of COVID patients when discussing booster vaccines for hospital workers: "I'm a healthcare worker and I don't take care of COVID patients specifically, but there can be COVID in the hospital, obviously. My patients are neurosurgery patients, so it's a little different" ("Gupta" [00:59:20 – 01:00:18]). In this sense he seems similarly situated to Dr. McCullough as a cardiologist, as it is unclear that he directly sees COVID patients either. However, unlike Dr. McCullough who still demonstrates with powerful anecdotes his healing orientation towards COVID patients, Dr. Gupta does not and this weakens Dr. Gupta's status as an expert.

Further weakening his embodiment as a physician, and therefore the embodied capital which would support his expertise, is that his role as a CNN correspondent, not his role as a physician or neurosurgeon, became the central object of focus in the podcast episode. Dr. Weinstein too, can be conceptualized in the journalist role. He does not invoke any of his own scientific endeavors in the podcast but instead hosts a Youtube podcast called the “Dark Horse Podcast” about COVID. Nevertheless, even Dr. Weinstein implies a direct relationship between his work in evolutionary biology and COVID, explicitly stating, as previously mentioned, that “evolution is the kind of good generalist toolkit to apply to something like COVID” (“Weinstein & Kory” [00:02:15 - 00:02:39]). On the other hand, while Dr. Gupta, briefly mentions his field's applicability to COVID, he fails to make the connection explicit: “I’m a neurosurgeon. When I first heard that this respiratory pathogen could cause isolated loss of smell, I thought to myself, like, the fact that they can't smell...how is it doing that?... Why is it causing strokes in young people? Why is it causing all these other organs to have problems?” (“Gupta” [00:36:41 – 00:37:22]). Briefly referencing “strokes” gets at the unknown, neurological symptoms of COVID, but it is easily missed as he does not articulate to Rogan the ways in which his skills and competence as a physician and neurosurgeon are relevant to these phenomena. Instead, he refocuses attention, concluding “You know, it's really weird. And this gets back to the novel nature of this virus. We don't know a lot about what this virus does to the body” ([00:37:22 – 00:37:44]). Whereas the other experts rhetoric center their active fulfillment of the role of physicians and scientists to aid patients and make progress, he presents a narrative of uncertainty. While this rhetoric is more scientifically accurate, it is less rhetorically palatable. Dr. Gupta’s failure to embody the role of the expert in the same way as the alternative-experts speaks is why his expert status is comparatively weak.

Though many would object to positioning the alternative-experts over the mainstream-experts and specifically over Dr. Gupta when it comes to expert status, the fact that Rogan's followers do so demonstrates the deeply rhetorical nature of expertise and how its perception does not always align with what is considered the ontology of true expertise. The symbolic capital of expertise is ultimately aligned with an ideal of what a biomedical expert is, and thus any given real figure must align with that rhetoric to be perceived as an expert.

Yet reproducing the individual markers of biomedical expertise is only half of the rhetorical work of gaining insider status. While the symbolic capital of degrees and embodied culture establishes the alternative-expert as a credible individual, there is also a rhetorical salience in being placed within communal structures through which biomedical knowledge is collectively produced and validated. The next section examines how the alternative-experts situate themselves within the culture of biomedicine, becoming recognized members of a scientific community and thus gaining expert credibility.

A Jury of Your Peers: The Rhetoric of Consensus

Alongside reproducing the symbols capital of science, I argue that situating oneself and one's claims within a community which validates oneself and one's work generates the insider status of biomedical science, which creates expert credibility. Alternative-experts establish themselves, their work and their claims as rooted deeply within a constructed scientific community, appealing to the ideal of established science resulting from a communally validated knowledge-making practice.

Generally speaking, communal validation validates the speaker. The social proof effect explains a general desire for agreement and validated competence, which is why some measures of peer support and community are important in establishing expertise. Social researchers Brand et al's research on the prestige bias, when people are copied by or listened to by others extensively, demonstrates the role of agreement in propping up knowledge-making (Brand et al.).

However, an ethos of communal validation is especially important in biomedicine because of the culture of professionalization and the role of consensus in science. Communal validation functions as a mechanism of proof in the culture of the biomedical community, thus, situating oneself within this culture is necessary to prove insider status and biomedical expertise. Returning to Starr's characterization of professionalization and the authority, legitimacy in the medical profession comes in part from how the "knowledge and competence of the professional have been validated by a community of his or her peers" (15). This demonstrates that individual practitioners' knowledge must adhere to communal claims as a whole. Moreover, consensus serves as a validating mechanism for scientific knowledge, which is why embodying the specific mechanisms of communal validation in the sciences are rhetorically useful. Ranalli, discussing how perceptions of hard-won consensus are important in shaping climate skepticism and

mainstream thought, writes that consensus is not just agreement, or an “artifact of groupthink, or interest, or bias,” but instead “emerges only after vigorous debate and a thorough examination of the range of alternative explanations. It is one in which centrifugal tendencies are strong, and the experts are drawn into agreement only reluctantly and after careful consideration”-- these mechanisms of consensus are important because it “provides some assurance that the scientific community has done its due diligence” (Ranalli 187-188). In summary, the consensus based communal validation of consensus is not simply agreement, but a potent form of evidence in and of itself within the broader biomedical community. For these reasons, relying on communal validation becomes a vital way to reproduce the broader culture of science, and thus posit oneself as within it and adhering to its standards.

This communal situation functions by the invocation of peer-review papers, citation of other putative experts who support one’s claims, and situating oneself within the communal dynamics of professional/scholarly groups and even governmental groups. Though alternative-experts would appear to be a complete “outsider” of the scientific community due to their disagreement with the mainstream consensus, they skillfully make claims and then construct a parallel, alternative-scientific community around themselves in which their claims are communally validated. Difficulty in differentiating between the alternative-experts scientific community and the mainstream-experts scientific community allows insider status to persist. This section demonstrates how consensus and involvement within the community of science are rhetorically invoked, creating insider status and lending credence to these figures’ expertise.

Situating Their Work Within a Community

The alternative-experts situate their claims within a communally validated scientific discourse primarily through the strategic invocation of peer-reviewed literature: a rhetorical

move that reproduces the appearance of the consensus-building process through which scientific knowledge is legitimated. Consider Historian of Science Naomi Oreske's appraisal of the legitimating mechanisms of science, that "every scientific paper and report has to go through peer review... no claim can be considered valid until it has passed peer review." (Merchants of Doubt, p. 17). This quote explicitly creates a standard of communal validation, literally from "peers," to justify new claims. Papers become a representation of the culture of communal validation, because of the standardization and mandatory peer review within them, thus they rhetorically function as the highest form of peer validation. The alternative expert's production of peer-reviewed papers, thus allows for their work to be communally situated. For example, consider Dr. Kory's discussion of the hospital protocol and the paper he published on it. A Google search reveals that he actually published three papers on it, each of which were peer reviewed, furthering his scholarly involvement. Although writing papers and conducting research is considered a contributory ability, it requires and implies high levels of competence from the part of the researcher in that field, and well as from the peer. Moreover, the papers imply communal validation, as they must be accepted by peers to be published. Therefore, the alternative-expert's production of peer-reviewed scientific papers thus strengthens a viewer's perception of their expert status.

Aside from their own direct research, citing the research papers and work of others that agree with them strengthens their communal positioning and thus their expert status. It allows them to counter a predominant claim which the mainstream levies against them: that their scientific evidence is flawed, cherry picked, or does not reach a critical mass which would validate their ideas. As the section on embodied academic capital demonstrates, these papers are useful in proving their engagement with the tools and mechanisms that create the embodiment of

biomedical expertise. However, in this light one can see that the papers not only speak to their individual scholarly activities and embodiment, but situate them within a wider community where multiple other experts (seem to) agree with their claims. To review the many papers invoked in the embodiment capital section in this new light, consider the following: Dr. Kory claims that “a famous French toxicologist reviewed 350 studies on ivermectin. And, uh, he was contracted to do this and he put his report out about a month and a half ago. And in the executive summary, he writes that severe side effects from ivermectin are unequivocally and exceedingly rare” (“Weinstein & Kory” [00:54:20 – 00:55:06]). The mass of papers, 350, in and of itself implies a critical mass of scholarly, peer reviewed work has been created for ivermectin. Moreover, invoking the role of a toxicologist implies credibility through academic credentialing, as toxicologists typically have advanced degrees in their field, at least an M.S. The word “famous” further seeks to improve credibility, and the stem of the word, “toxic,” implies there is a match between field of expertise and field of study (safety of a drug), further contributing to something which increases expert status. Situating their claims as supported by a broader culture of peer reviewed literature implies communal validation of the work, and situates the alternative-expert himself within the broader consensus-mediated values of the field.

Aside from invoking general papers and studies, there is a frequent reliance on population level studies, which both emphasizes the credibility of the papers by situating it on a large scale and implies that the study provides quality evidence because of the sample size. Dr. Kory explicitly makes the appeal for why this form of evidence is credible, stating “you're also getting data from [the] real-world. That's a really credible, [pause] in fact to me, that's probably the most powerful source of evidence as you're seeing it work on a population-based, on-on a population basis around the world” ([01:28:47 – 01:28:50]) Indeed, large quantities are often used as a

heuristic to imply quality. For example, Dr. McCullough claims that Mexico “implemented a nationwide test and treat program. Every outpatient testing center, if you tested positive, you were offered Ivermectin...within two weeks of that, hospitalization rates plummeted, death rates plummeted. And over the next three months, they basically rid COVID, opened bars, opened restaurants at a time when the vaccination rate was like one to 5%. So it wasn't the vaccines, it was all related to this.” Notably, this isn't just any paper, the subject who organizes this study is “Mexico,” implying legitimacy through both scale and governmental authority. Thus, invoking papers and publications not only appeal to the internalization of the scientific community, but also situates the expert within the communal validating mechanism of the function as a communal validator because of the peer-review process, and papers are especially important as the currency of scientific discourse, allowing these experts to situate themselves within a seemingly communally validated scientific discourse.

While the mechanisms of peer review are implicit, and that not all viewers may recognize the enforced process of communal validation, the alternative-experts explicitly discuss how other experts have validated the papers, emphasizing and making visible the role of communal validation, even if one does not know the specifics of papers. Dr. Kory, invokes the role of others in a recently published study to support his work on the hospital protocol, stating a study was,

“just publish[ed] this weekend by...Andrew Bryant, Tes Lawrie, [and] Scott Mitchell, actually, who is a member of the FLCCC¹⁰... [they are] a group of researchers who for decades, their main job is to review medicine-- medical evidence to formulate guidelines for the big national [and] international healthcare agencies. Let's go back to that term I used before, they did this work independently. No one paid them to do this work. They did it because they saw-- they looked at my paper and they saw my testimony, and they

¹⁰ The FLCC is an organization that Kory is a part of, so the claim of independence is dubious. However, at this point in the episode, the FLCC had not been brought up: about 20 minutes later, it was introduced. It is unlikely that anyone watching this episode for the first time would make the connection that the acronym that was cursorily mentioned was the same. So, in this instance, invoking the FLCC solely functions as a mechanism of establishing expertise through involvement with professional organizations. As the next section will discuss, communal organizations allow the alternative experts to not just situate their claims, but their broader expert status, within a community that validates them.

immediately got interested and they started researching and they found consistent positive reproducible segments.” (“Weistein & Kory” [00:54:20 – 00:55:06])

Dr. Kory relies on the power and responsibility of these individuals which is implied by defining their role as “formulat[ing] guidelines for the big national [and] international healthcare agencies,” and the expertise implied from “decades” of experience. Those with power over such high-level guidelines are viewed with a high level of power and thus expert credibility— similarly to the aforementioned figures who seemingly determine hospital treatment policy. Moreover, emphasizing the author’s financial independence removes doubt in their motives for publishing the paper and confirming Dr. Kory’s findings. He claims that the meta-analysis finds on average a “62% reduction in death when you use ivermectin from all of these randomized control trials¹¹ ([00:55:10 – 00:55:20]). Not only is he invoking a paper which supports his work- the meta-analysis recently published- situating his research and claims within a scientific discourse which confirms his findings, but in elevating the expert role of the authors of that paper and crediting them with pure morals, his own expert status is increased.¹²

It is important to consider that while communal validation through the research of others is rhetorically employed by both the mainstream and alternative-experts, the clear science communication of the alternative-experts seems more effective when contrasted with murky and

¹¹ A google search of the author’s names and “ivermectin meta analysis” brings up the paper titled: “Ivermectin for Prevention and Treatment of COVID-19 Infection: A Systematic Review, Meta-analysis, and Trial Sequential Analysis to Inform Clinical Guidelines,” which was published June 21st 2022 (the day before the podcast episode, so it would be available to viewers), which does conclude the 62% reduction deaths, lending credibility to Kory’s claims.

¹² Dr. Kory is not the only figure to invoke this: Dr. McCullough both heightens his own and other scientist’s expert credibility who concurs with his myocarditis claims, stating “we’re both graduates of the University of Michigan, which by the way is, you know, I think it’s one of the better places in the United States. He went to medical school there. I want to graduate school there. I went to UT Southwestern. I finished top of my class. I’m an Alpha Omega Alpha. The doctors who were in the know of how to treat COVID-19 were no chump change” (“McCullough, [02:34:50 - 02:35:07]). By emphasizing the prestige of these academic institutions, the accomplishments of the physicians (graduating “top of my class”) alongside explicitly stating: those who are “in the know” on COVID treatments are “no chump change” emphasizes both McCullough and his colleagues academic qualifications, and places himself within a community of credible experts.

confusing claims made by the mainstream-experts. For example, a lot of the mainstream efforts to counter ivermectin usage in COVID claimed that ivermectin was not safe, but this argument was seen as fraudulent due to the oversimplification of this talking point. The claims that ivermectin is not safe if taken without a prescription are reasonable, as self-administration of a drug without a prescription from a physician can lead to dangerous consequences due to overdosing (“Pharmacist Details Ivermectin’s Dangerous Snowball Effect”). However, this point is often oversimplified to some formation of *ivermectin is not safe*, which is easily contradicted by Dr. McCullough’s claims that Ivermectin is “safe and effective. [There’s] been a Nobel Prize awarded in 2015 for Ivermectin” (“McCullough” [00:25:36 – 00:29:02]). The Nobel Prize, is one of the most prestigious and well-known awards, and selection is by peers, implying communal validation. It seems unbelievable that a drug which won the Nobel prize could also be unsafe. Similarly, Dr. Kory draws out this “weird behavior” in advice on Ivermectin usage, stating that the WHO simultaneously questions the safety of Ivermectin for COVID on their page, but simultaneously, “there are other guidelines for the other diseases that ivermectin is [used for] from the WHO ...they’ll write in [those guidelines] that billions of doses [have] been administered, the side effects are minor and transient” (“Weinstein & Kory” [00:56:20 – 00:56:54]). This contradiction raises doubt on the accuracy of the claims by the WHO, but the complexities of why this contrasting information can be true as explained above are not presented, and thus, the complexity becomes evidence of a lie. Altogether, it seems like an unresolvable paradox: ivermectin is either safe or it isn’t. The alternative-experts’ mobilization of communal validation of ivermectin seems to lend credibility to the former claim.

However, it is not only the situation within communal validation that accomplishes this—this misconception is able to persist because the alternative-experts cite others who do the same

research and find the same conclusions, independently, which validates their own expertise and findings while supporting the creation of an alternative academic discourse.

Situating Themselves Within a Community

Alongside situating their claims within a community of validated knowledge, the alternative-experts must also situate themselves within a community that validates them as experts. Where Section 1.2.1 focuses on the communal legitimation of their claims, Section 1.2.2 examines how organizational memberships, invocations of prestigious peers, and affiliations with powerful institutions work to suggest that the alternative-experts themselves have been recognized, and therefore belong, within the insider communities that confer expert status. When it comes to peers, other scientists and physicians take primacy in validating claims. As Starr writes, “Medicine and other professions have historically distinguished themselves from business and trade by claiming to be above the market and pure commercialism. In justifying the public's trust, professionals have set higher standards of conduct for themselves than the minimal rules governing the marketplace and maintained that they can be judged under those standards only by each other, not by laymen...” (23), demonstrating the historic roots of communal validation as a mechanism sustaining professions.

Though the mechanism itself is internal validation, this validation can be perceived from an outside vantage point as a layperson. One mechanism by which this broad, character peer validation is perceived as granted is through involvement in professional, governmental, academic, or honorable scientific societies. These memberships contribute to the construction of an alternative community because they rhetorically imply that the other members of that society have at least a base level of agreement with the physician or scientist's work and claims. The validation of their work is evidence of their expertise. Philosopher of science Neil Levy

conceptualizes this phenomenon by understanding these organizations¹³ as a “mimicry of [the] markers” of expertise, which artificially “inflate the appearance of expertise” (Levy 112)

Alongside serving as a mimic, the creation of alternative groups and institutions allows for new structures of communal validation to emerge. This alternative-consensus both creates uncertainty in the consensus of the mainstream body and attempts to supersede it. However, these organizations can mimic mainstream communal validation without actually representing it. In the book *Bad Beliefs: Why they Happen to Good People*, philosopher Neil Levy demonstrates how bloated the landscape of professional organizations has become and how this makes examples of communal support abundant, using the example of the American College of Pediatricians (ACP). This organization released a statement condemning gender reassignment surgery and calling it “child abuse,” and many mistook this statement as representing the much larger American Academy of Pediatrics (AAP) or the views of pediatricians in general (LaCapria) (Levy 112-113). Though it is not representative—consisting of approximately 700 people as of 2023, ((Cameron and Mehrotra), whereas as of 2026 the AAP has about 67,000 members (“About the American Academy of Pediatrics”), the confusion still persisted due to the name of the organization and general lack of understanding of the size of its membership. This example demonstrates how communal validation can be mistakenly perceived as mainstream communal validation.

This dynamic of implied communal validation through involvement with organizations is replicated in the Joe Rogan Experience podcast. For example, referring to a public symposium he attended alongside 500 other doctors who are alternative-experts, Dr. McCullough states “Many of us are members of the Association of American Physician and Surgeons, or the Frontline

¹³ Alongside entities like journals and websites which also parallel mainstream structures and institutions, as well as attempts to reproduce a great number of the markers of expertise, such as the educational credentials and research experience discussed in the chapter.

Critical Care Consortium, or American Frontline Doctors, or the Truth for Health Foundation” (McCullough, [02:23:56 – 02:26:33]). His involvement in so many societies situates his person and his claims within numerous communities of physicians. Since involvement in these societies implies a base level of support and agreement, this societal involvement creates communal validation which bolsters his expertise.

Typically, these organizations possess authoritative and general names which can evoke authority while also leading to conflation with other, mainstream organizations which they parallel. Specifically, Dr. McCullough’s involvement with “The Association of American Physicians and Surgeons” (AAPS) creates the opportunity for a conflation with the much larger and more mainstream organization- the “American Association of Physicians” (AAP). In reality, the ideas supported by the AAPS are quite distinct from the AAP. The AAPS is relegated to the fringes of mainstream scientific thought because of its HIV-AIDS denialism, climate change-denial, support of the abortion-breast cancer hypothesis,¹⁴ as well as support for a connection between autism and vaccines. Despite the differences in content and ideology, the similarities in names allow for these organizations to be conflated, as with Levy’s example of the ACP and the AAP. These organizations present alternative ideas through a lens of group validation, creating an “alternative-consensus” which mirrors the mainstream, but their conflation with mainstream bodies allows their positions to both be interpreted as mainstream and also to attempt to supersede the mainstream consensus.

Within these organizations, the alternative-experts often present themselves as leaders; The position of these individuals as experts within these organizations improves their expert status by implying not just general peer validation, but a higher level of recognition that comes along with leadership. For example, Dr. McCullough states that he is the “editor of a major

¹⁴ Which states that abortion increases rates of breast cancer

journal in cardiovascular medicine, the former editor of an international journal, and the president of a major medical society...about five years into that position” alongside a separate mention of having had “a very highly ranked position at a major academic medical center” (“McCullough” [00:02:04 – 00:03:55]) While journals are not strictly professional or honorable societies, the status of publications creates an air of importance around editing a journal, and thus implies a similar recognition through leadership as the other organizations and the medical society that Dr. McCullough references. The repeated use of the words “major” and international further increases his expert status.¹⁵ Leaving out the names of the organizations makes it difficult to verify institutional involvement or the credibility of the institution, allowing perceptions of communal validation to exist where they otherwise may be countered. Nevertheless the communal validation from his organizational involvement and leadership role in these communities increases his expert status.

Alongside invoking their involvement with other organizations, they situate themselves within the mainstream scientific community by invoking other figures who fulfill the expert role through educational and academic criteria. For example, how Dr. Kory contextualizes his “Frontline COVID-19 Critical Care Alliance,” an organization he is involved with, as communally validated by invoking another putative expert. Within the podcast episode, Dr. Kory describes the formation of this organization during the pandemic when “Dr. Paul Marik, a very famous guy in our specialty....[the] most published intensivist in the world...” was approached by “people [who] came to him to develop protocols” on COVID. So, “[Marik] grabbed his-his four closest colleagues and friends,” including Dr. Kory, and they “developed the hospital

¹⁵ I would like to note here that I, too, was the editor of a national journal. I mean, it was my high school’s poetry and literary journal, and it was distributed both in North Carolina and South Carolina because some of my peers had family in SC and took it to them... but still, that counts as “national,” right? Even so, I was specifically chosen to be the editor of this fine publication (the last editor was a close friend of mine, and nobody else wanted to do it, but nevermind that), and that surely signifies my status as an expert in the literary world!

treatment protocol back in March [2021]” before shifting their focus to ivermectin treatment protocols (“Weinstein & Kory” [00:03:05 – 00:04:57]).

Dr. Marik himself fulfills the role of another expert, which increases the credibility of the organization and thus Dr. Kory as an expert. Dr. Marik is described as the “most published intensivist in the world,” again utilizing the heuristic of large numbers and invoking publications to establish Dr. Marik as a leading expert. Having established Dr. Marik as a leading expert, the group he organizes gets a boost in status. Moreover, the professional (and interpersonal) respect Dr. Marik shows for Dr. Kory increases Dr. Kory’s status as an expert as well. It seems only logical that if a leading expert scientist trusts another scientist for their scientific work, then the latter scientist must at least somewhat know what he’s talking about as well.

Aside from Dr. Marik, an amorphous “other” is invoked who is imbued with a large amount of hospital policy power, and this raises the credibility of Dr. Marik the organization, Dr. Kory, and all others involved. Specifically, Dr. Kory says that “people came to Marik” to ask him to develop a protocol which is then labelled by Dr. Kory as “the hospital treatment protocol.” By referring to it as “the” hospital treatment protocol, there is a suggestion that it applies to all hospitals across the nation. Similarly, the use of the vague term “people” who approached Dr. Marik suggests that he had been selected by a large body of individuals with decision making authority and given a great deal of responsibility. It implies that Dr. Marik had enough of a successful track record so that those who influence national hospital policy handpicked him for this vital task. This invocation of a vague other and Dr. Marik allow for Dr. Kory, his organization, and his work to be situated within a community which not only validates but lends incredible recognition to their biomedical work and therefore their expertise.

The role of the powerful “other” also demonstrates that situating oneself within any powerful community, not just one focused on scientific discourse, increases the status of these alternative-experts. The alternative-experts situate themselves within networks from the government and others in power, creating alternative sources of credibility with which to situate themselves in a community which validates their expertise. Consider Dr. Malone, who similarly to Dr. Kory and Dr. Marik, was seemingly approached by powerful others who were in need of his research skills, but in Dr. Malone’s case, it is security agencies in the US who approached him:

“there's a CIA agent that I've co-published with in the past named Michael Callahan...he called me from Wuhan on January 4th I was currently managing a team that was focusing on drug discovery for organophosphate poisoning ergo nerve agents for DTRA, defense threat reduction agency, involving high-performing computing and biorobot screening um high-end stuff and he told me: ‘Robert you got to get your team spun up because we got a problem with this new virus,’ I worked with him through prior outbreaks and so it was then that I turned my attention to [COVID]...” (“Malone” [00:20:06 –00:22:00]).

Invoking both DTRA and the CIA, US government organizations with seeming immense power over national security,¹⁶ increases his desirability as a researcher. As I will discuss in Chapter 2 , the government can be cast as corrupt and thus untrustworthy, yet specific invocations of defense agencies appeal to the nationalist sentiments of our time, allowing them to circumvent these concerns.. The scientific jargon and explicit reference to a vague but important qualification of items as “high-end stuff” further inflate his abilities as a researcher, while a reference to working with Callahan through “prior outbreaks” improves Dr. Malone’s status as a pandemic/ outbreak expert. The dialogue sounds straight out of an action movie, which combined with the invocations to powerful government agencies increases Dr. Malone’s status as

¹⁶ This is widely recognized in the case of the CIA. DTRA is likely less familiar to the American people but appears similarly official and important based on Dr. Malone’s description.

not just a desirable researcher, but a hero figure. Both his desirability as a researcher and this character trait improve his putative expert status.

Why Communal Invalidation is Ineffective

If peer validation situates claims and persons within an expert community, and thus improves their claim to expertise, it reasonably follows that peer invalidation should detract from expert status. There are plenty of examples of the mainstream-experts condemning the alternative-experts and their claims. For example, when 270 experts wrote to Spotify to protest Dr. Malone's appearance on Rogan's podcast. Nonetheless, Dr. Malone's expert status persists. I argue that this demonstrates that something more complicated is going on.

There are multiple reasons for this persistence despite invalidation by the mainstream-experts. The first is due to the difficulty of debunking these scholar's claims, as they are actually difficult to address. The second reason is due to the temporality of the debunking, and the third reason is because of media positionality. Echo chambers make it so that some content is more likely to be shown to us, or accessible enough for us to access on our own (Di Martino et al. 1). Because of this, many debunking efforts are neither viewed, nor integrated. Fourth and finally, even if the information does reach us, after ideas become established, it is simply difficult to challenge them..

Starting from the first, the selective vagueness of Dr. Kory's description of his hospital protocol continues to fuel this organization's and his own expertise by making it difficult to verify the success of the protocol or the broader validity of his statements. The first barrier to verifying the information presented is that Dr. Kory does not name the protocol itself. Viewers who are looking to validate his claims would have to turn to another source— in the context of a podcast episode, which is often listened to while driving, working out, or doing other work, rates

of investigation are likely limited. Similarly, it is difficult to conclude exactly who the “people” are. Are they perhaps hospital administrators? High-up physicians who make policy decisions? Without knowing who they are, it is nearly impossible to determine if their implied power over the healthcare system is real. The nature of their seemingly flattering interactions bolsters Dr. Marik and therefore Dr. Kory’s expertise. Moreover, those who do seek to learn more information and verify Dr. Kory’s claims will perceive him as a credible source. A simple Google search for “Kory Hospital Protocol” on Google yields three papers about a “MATH+” hospital COVID protocol based on repurposing drugs such as Methylpridione, Ascorbic acid, Thiamine, Heparin and more (including ivermectin and famotidine) to treat COVID, and Drs. Kory and Marik are listed as authors. Moreover, invoking these papers– the mainstream currency of science– demonstrate another way in which the alternative experts gain status: the alternative experts bolster their claim to expertise by situating not only his person within an organization or associating themselves with other perceived experts, but by situating his work within the broader academic discourse of science.

Mainstream experts would likely object that one of these papers was later retracted, and other mainstream-experts would criticize and poke holes in the findings; this serves as a form of communal invalidation, which should have the effect of lowering the alternative-experts perceived expert status. However, it is important to consider the timeline in which these criticisms became available, as first impressions in viewer perception of these figures causes subsequently presented information to become less impactful on viewer perception. First impressions tend to stick around. One of the three papers was ultimately retracted after the Sentara Norfolk General Hospital in Norfolk, Virginia, where the study was conducted, raised

concerns about the validity of the data presented: As reported by the blog Retraction Watch, Sentara found that mortality reports for the MATH+ interventions were inaccurate (Marcus).

However, the availability of this information does not mean it would be integrated into people's perceptions of Dr. Kory's expertise. However, Dr. Kory appeared on the podcast in June 2021, and the retraction was in November 2021, so this article would not have been labelled as retracted in case JRE viewers decided to look it up for themselves at the time the podcast was published.

Beyond temporally limited access, the position people occupy in their social media networks can render seemingly accessible information out of reach. In the age of social media, it is increasingly important to consider the role of algorithms and echo chambers, which makes it so that different communities will see different content, and thus, have different data and views of the world presented to them (Di Martino et al. 1). At the time of the podcast episode airing, there was an article called "Doctors publish paper on COVID-19 protocol; Experts Unconvinced" which criticizes the validity of Dr. Kory's protocol. This article was published in *The Hospitalist* and *Medscape* in December 2020 and thus would have been available to the listeners of Kory & Weinstein's podcast episode at the time it aired for verification purposes ("Doctors Publish Paper on COVID-19 Protocol; Experts Unconvinced"; Hickman). Moreover, this article is written in plain language, which increases its accessibility to laypersons, and allows them to interact with mainstream-experts and perceive the communal/peer-invalidating of Dr. Kory from mainstream-experts, thus lowering his expert status. Yet these forums are academic in nature: Medscape's about page lists that their "mission is to improve patient care with comprehensive clinical information and resources essential to physicians and healthcare professionals," demonstrating an orientation towards reaching non-layperson biomedical

practitioners (*About Medscape*). On the other hand, YouTube, a media hub geared toward the general public, had a video titled “MATH+ Treatment Protocol for COVID-19 Explained!” available at the same time- as it was published June 8th 2020 on a channel which describes itself as “educational” and had 937K subscribers (as of 2026). As the Joe Rogan Experience podcast is hosted on YouTube (amongst other platforms), viewers are more likely to be familiar and look to YouTube for similar content as opposed to engaging with unfamiliar forums not marketed towards them. This demonstrates how in a media age, discovering dissenting pieces becomes difficult considering the variety of forums that we selectively engage with. In Dr. Kory’s invocation of Dr. Marik, these alternative-experts invoke other figures, quickly define them as experts in their own right, then show that these figures support their claims and expertise. This situates their research within a broader community and their association transfers respect, which both serve to increase their own expert status.

As previously discussed, the timeline where information which would have invalidated or sewn doubt in Dr. Kory’s claims is important to consider, as the audience’s perception of his expertise is increased when they do not have access to information which would invalidate him: Though the meta-analysis, to this day has not been retracted, it is important to note that meta-analysis do not collect their own data, they look at other studies, such as randomized control trials, and seek to draw conclusions from combining the data from all those studies. This particular meta-analysis used data from 24 randomized control trials (aka, trials which give an intervention randomly to one group and give nothing or another form of control to another group). Two of the included trials, Elgezzar et al and Niae et al, were eventually retracted because of concerns about fraudulent data. While 2 out of 24 articles does not seem like it should significantly influence results, the Elgezzar study represented “10% of the effect size” of the

meta-analysis, and those two studies are the only ones which independently concluded that ivermectin reduced mortality (Lawrence et al.) Indeed, this prediction was fruitful, but this was only confirmed a year later. Another author, Hill et al, initially conducted a meta-analysis using both Elgazarr and Niae's papers, concluded alongside Bryant et al that ivermectin reduced mortality on July 6th, 2021 (Hill et al., "Retracted"). On August 9th, Hill et al published an expression of concern on their own paper, indicating they had been made aware that one of the studies in the meta-analysis was fraudulent (Hill et al., "RETRACTED"). They retracted the article February 5th, 2022 and in January/ February their new article "Ivermectin for COVID-19: Addressing Potential Bias and Medical Fraud" appeared, showing that when the "high risk" for bias and potentially fraudulent studies were removed (Elgezzar, Nica, and two others also included in the Bryant et al study) not only did mortality reduction rates drop to 10%, but the results were no longer statistically significant: Statistical significance is defined as an r^2 value less than or equal to .05 (scale of 0 to 1). The r^2 value was .66: for highly technical reasons, statistical significance is important because when an r^2 value is higher than .05, it is incredibly likely that the result was a result of random noise in the data, not an underlying cause (Lawrence et al.). Simply put, .66 is very far from .05, so this is not a statistically significant result, so the conclusion that mortality rates are 10% (still a low impact in and of itself) are likely entirely random. However, this paper appeared many months later, when the viewership's internalization of Dr. Kory's claims had likely already been formed.

Even if corrective information eventually reaches viewers, which as discussed, is unlikely because of echo chambers, its impact would be limited. Psychologist Stephen Lewandowsky and colleagues demonstrate in their article "Misinformation and Its Correction: Continued Influence and Successful Debiasing," that initial exposure to false or misleading claims produces a

"continued influence effect," in which the original information continues to shape beliefs even after explicit correction (Lewandowsky et al.,). The initial, convincing invocation of communal validation is likely to be far more rhetorically powerful than any subsequent debunking, as it arrives first, and shapes the interpretive frame of information gained later– if that information is gained at all.

In this way, communal invalidation comes too late to impact the viewership's perception of the alternative-experts communal validation and insider status of expertise.

The Inside Man: Conclusion

Taken together, the two rhetorical strategies examined in this chapter—the reproduction of symbolic capital and the situating of claims and persons within a communally validated scientific community—work in tandem to construct the alternative-experts as biomedical insiders. By invoking academic degrees, embodied clinical and research experience, peer-reviewed literature, and organizational membership, these figures reproduce the cultural imaginaries of what a biomedical expert is in the mind of the public, creating an insider status which increases their credibility. This allows them to be perceived as such by lay audiences regardless of their relationship to the mainstream scientific consensus. As the failure of Dr. Gupta's expert appeal demonstrates and the difficulty in disproving their communal validation shows, this perception is ultimately rhetorical: the mainstream-expert is equally reliant on reproducing these cultural markers, demonstrating how expertise is ultimately rhetorical.

Ultimately, this insider status remains the foundation through which all claims to expertise are ultimately made, because appealing to the cultural imaginaries of science ultimately creates expertise. However, the next chapter will explore how insider status is complicated and recast in populist and democratic contexts, creating a salient appeal that both relies on the broader cultural imaginaries of science and taps into specific ideologies that underlie the current US political context.

Chapter 2: The (Faux) Outsider: Recasting Insider Expertise with Populism

The idea of rhetorical outsider expertise is not new to vaccine-hesitancy scholarship. In 1998, Dr. Andrew Wakefield, a UK physician, published a case series in the *Lancet*, which suggested that the measles, mumps, and rubella (MMR) vaccine was linked to the development of the symptoms of autism. This study, despite being retracted, has had lasting impacts to this day, and Dr. Wakefield himself has gained a position of prominence within vaccine hesitant institutions. As Goldenberg writes in her book *Vaccine Hesitancy: Public Trust, Expertise, and the War on Science* (2021), “Despite persistent attempts by the medical establishment to discredit him in an effort to calm public fear, Wakefield has a dedicated following and continues to influence public response to childhood immunization” (152). She characterizes his appeal as “exemplary of the maverick—the free-thinking and unorthodox eccentric,” which she further characterizes as an archetypical scientist who “wield[s] outsider expertise” and “powerfully counters the once-presumed warrant of expert-driven decision-making on scientifically informed policy” (153). In other words, the maverick’s appeal lies in his “outsider” character and both starkly contrasts and raises questions about the mechanisms of expertise and scientific decision making as it applies to the public. This demonstrates the importance of outsider status in the perception of expertise in vaccine-hesitant scholarship.

The alternative-experts rely on similar outsider appeals as Goldenberg’s maverick archetype, which increases their rhetorical appeal as experts. However, Goldenberg’s conceptualization of the appeal of outsider expertise sacrifices the role of inside status as constructing expertise. She writes, “[t]he perceived trustworthiness of the maverick lies in his distance from organized science; this contradicts Anderson’s hierarchy, in which near proximity

to the establishment's core ensures credibility" (163). Anderson's hierarchy, which Goldenberg draws on, situates scientists with good academic credentials, such as being trained in the field, being leaders in the field, well cited and recognized by peers over laypersons and those with only tangentially related academic experience (see table 1).

HIERARCHY OF EXPERTISE
Scientists who are leaders in the field
Scientists whose research in the field is recognized and cited by peers
PhD scientists trained in the field
PhD scientists outside the field, but with collateral expertise (e.g., statistics in the relevant field)
PhD scientists outside the field of inquiry
People with a bachelor's or professional degree in an applied science specialty far removed from the field
Laypersons

Table 1: Anderson's Hierarchy of Expertise, reproduced from Goldenberg's book for the purposes of understanding Anderson's claims(161)

Yet Goldenberg's argument, that the maverick flips this hierarchy, is inconsistent with what I demonstrated in Chapter 1. Firstly, I showed that alternative-experts fulfil the criterion of being trained in their field, with the use of degrees as institutional capital and the close mapping between experience and claims to expertise in Section 1.1. Secondly, I showed that alternative-experts are recognized and cited by peers, demonstrating their place within the biomedical community in Section 1.2. Finally, their leadership status is demonstrated throughout Chapter 1 and in Section 1.2.1, as they produce research and become leaders of groups which confers upon them additional expert status. Utilizing all of these methods, alternative-experts rhetorically reproduce the insiders claims of expertise.

In this way, the claim that the alternative-expert solely benefits through their outsider status, and through flipping the hierarchy of traditional expertise, are too conceptually simple to explain the rise of this new type of expert. Insider status, recast through populist appeals, resembles outsider status, but is really just variations of insider status repositioned with a transcendent distance. As a result, I refer to this phenomenon as *faux outsider status*. This coalescence of inside and outside status emphasizes that expertise is not constituted from considering each status separately, but instead comes from combining both to create something new. The careful maintenance of a multi-faceted status creates a poignant form of expertise.

Understanding the metanarratives, values and implications of populism and democracy is necessary to understanding how expertise is recast through them. Populism, at its core, is a battle between *the people* and *the elites* (Meyer 846). Political Scientists Jane Mansbridge and Stephen Macedo, apply these ideas to form a conceptualization of Science-Related Populism in their article “Populism and Democratic Theory,” arguing that populism “pit[s] the people in moral battle against elites,” while also tapping into democracy’s “normative roots in the wants and needs of ordinary citizens. . . challenging, on egalitarian and justice grounds, elite political, economic, and cultural domination,” demonstrating how science, populism and democratic appeals become linked together (Mansbridge and Macedo 60).

Indeed, further work shows how the role of the maverick becomes increasingly salient within populist cultural shifts. Robert Barr, for example, identifies the maverick as a central figure in populist appeals. In the *Journal of Party Politics*, he defines populism as “a mass movement led by an *outsider* or *maverick* seeking to gain or maintain power by using *anti-establishment* appeals and plebiscitarian linkages” (Barr 38,; emphasis added). Goldenberg explains how maverick characters, such as Dr. Wakefield, typify a populist representative of the

people: She explains that scientists are seen as experts in their own niche fields, and that this expertise is balanced with a democratic accountability to public values. Within this political context, she argues, the maverick functions as a “populist counterforce to the power interests that expert elites [ie, the mainstream-experts] are thought to represent and reinforce” with an “antiestablishment persona,” demonstrating how this archetype is situated within populist metanarratives of the elite versus the people. The maverick, thus, is a character who holds expert opinion accountable to public values, its appeal is particularly salient when considering democratic values. (Goldenberg 153).

This rise in populism is often identified with an outright rejection of expertise. In their book *Experts and the Will of the People: Society, Populism and Science* (2020), Sociologist of science Harry Collins and his co-authors note that "the rise of populism in the West has led to attacks on scientific expertise," arguing that "scientific expertise is one of the checks and balances in pluralist democracies and that is why populists attack scientific expertise," directly positioning populism as an antagonist to biomedical scientific expertise. Goldenberg's book, which argues against the rejection of expertise, gives a compelling example for how this theory of the “Death of Expertise” is cast under populism: “where once experts were valued for the utility that learned wisdom offered for solving complex problems, expertise has been succeeded by populist common sense (Goldenberg 73). These quotes demonstrate the widespread conceptualization of populism as antithetical to biomedical expertise.

While Goldenberg argues against this Death of Expertise, arguing instead that there has been a “*recalibration* of expertise by members of the public,” she mostly examines populism through the lens of the 'death of expertise' theory, treating it as producing a new form of expertise only insofar as its rhetorical appeal is rooted in opposition to mainstream science. This becomes

evident in her conceptualization of the maverick character, which is rooted in populism, but the expert character of this figure only appeals in a sense that it exploits the mistrust in the mainstream-experts as a source of credibility, not that it reproduces other forms of credibility on its own. Therefore, while she takes the crucial first step in arguing that the maverick represents a new form of expertise, she fails to comprehensively describe the rhetorical appeal that props up populist, outsider forms of expertise. This demonstrates the need for a new conceptual framework of outsider status, not just grounded in populist attack, but merging expertise and populist democratic values to form a new sense of credibility: not the maverick, but the alternative-expert.

Political Theorist John Meyer begins to take these conceptual steps in the context of climate change, recasting climate-science deniers as his own form of alternative-expert in his article “Power and Truth in Science Related Populism: Rethinking the Role of Knowledge and Expertise in Climate Politics.” He criticizes Mede and Schäfer’s Science Related populism framework as “inherently exclusionary and one that necessarily rejects expertise in any form, because it is perceived as a manifestation of elitism” as “foreclos[ing] recognition of the role played by other forms of knowledge—often characterized as counterknowledge or counterexpertise—within science-related populism” (Meyer 847-848). His criticism is postmodernism, recognizing the role of new forms of knowledge by invoking “counterknowledge” and “counter-expertise,” which provides a theoretical grounding to explore the rhetorical appeals to outsider alternative-expertise. Indeed, Meyer begins this work by conceptualizing a “lean into populist criticisms of elite expertise, by delineating how some challenges to dominant forms of science and elite power are themselves expert knowledge” (845). His work focuses on the consensus in climate justice. I seek to extend this, by drawing out

the various forms of rhetorical appeals which recast expertise under populist norms, and specifically extend it to a vaccine-hesitant context. Indeed, I argue that populism does not tear down expertise, which would be the simple conceptualization of outsider/ death of expertise theory, but rather recasts how expertise is understood.

The populist ideology of Rogan's audience aligns closely with this reading of Science Related Populism. As recent scholarship by professor of Science Communication Dominik Stecula on "Podcasting the Crisis of Masculinity: Recognition Politics, the Manosphere, and the Joe Rogan Experience Audience" demonstrates, Rogan and similar figures "frequently employ communication strategies common to populist rhetoric, notably anti-elitism, anti-intellectualism, and appeals to 'common sense'," which resonate with audiences who feel alienated from institutional authority (Stecula et al. 6, qtd Mudde & Rovira Kaltwasser, 2017). He continues, "This style effectively taps into feelings of disenfranchisement and marginalization among listeners, reinforcing narratives that elites— academic, political, or media—are disconnected from or actively hostile toward the lived realities of ordinary people, particularly young men experiencing social and economic uncertainties" (6). This anti-elite attitude actively demonstrates the ideology of the Rogan viewership, but also demonstrates how it operates in a larger social context of discontent, speaking to this ideology's widespread appeal. Stecula also writes that the podcast operates as an epistemic community shaped by what scholars term "epistemic polarization—" that is, "a divergence in what different groups recognize as valid knowledge and trustworthy information sources" (8). In this way, Rogan's platform becomes not just an alternative source of information, but a space where "shared skepticism toward established institutional narratives" is reinforced, making audiences "especially receptive to alternative viewpoints, conspiracy theories, and contrarian perspectives" (Stecula 8). Thus,

exploring the role of populism on the rhetoric of expertise in the Joe Rogan Experience Podcast becomes particularly telling.

Within this context, I identify three key rhetorical moves through which alternative-experts construct their complex insider outsider positionality. First, they establish a transcendent distance from mainstream institutions: rather than positioning themselves as pure outsiders, they invoke their insider credentials while simultaneously narrating a departure from the corrupt establishment, creating a tenable position of simultaneous proximity and distance that recasts what it means to be an expert under populist metanarratives. Second, they delegitimize the mainstream by casting it as corrupt, profit-driven, elite and antagonistic to the American people. This entangles pharmaceutical companies, regulatory agencies and academia in a web of self-interest that undercuts their epistemic authority. The scientific method itself becomes corrupt, being misused by malicious, corrupt actors and thus producing erroneous knowledge. With the mainstream delegitimized, something must fill their expert void. By positioning themselves as uncorrupted by profit and genuinely oriented towards the health of the people, as well as reproducing the insider values that constitute physician and scientist authority, they become appealing as experts. Third, they construct an alternative epistemology: rather than simply rejecting mainstream science, they recast how scientific knowledge should be produced and considered, elevating populist-friendly forms of evidence like anecdote and meta-analysis, democratizing the sites of knowledge production, and empowering the layperson as a knower in their own right. They do not simply tear down expertise, but rather recast it on populist and democratic foundations, creating a new form of credibility that is perceived as more legitimate than mainstream science.

Beyond the Maverick: A Transcendent Distance

Alternative experts construct a persona of brave dissent: enough of an insider to have credibility, enough of an outsider to tell the truth. Chapter 1 demonstrates the plentiful academic credentials of the alternative-experts, which set the stage for their authority and perceived expertise. Yet the alternative experts also differentiate themselves from the traditional role of the physician or scientist by creating a transcendent distance between their character and academia. Their expertise is grounded in an initial narrative of insider institutional understanding and belonging, followed by either a move to the outside or a criticism of these very same structures. This narrative of leaving appeals to populist distrust of academia while maintaining institutionally granted credibility. In this way, they are not just invoking outsider expertise as the pure outsider maverick, but creating a tenable position of simultaneous distance and proximity. This positioning fundamentally recasts what it means to be an expert, generating a new rhetorical appeal of an outsider with a transcendent distance, creating credibility in a populist political context.

A literal invocation of a journey from institutional belonging to leaving for a different field epitomizes the rhetorical salience of this new form of expertise. As described in Chapter 1, a variety of forms of academic credentials and academic clinical experience/ research are invoked. After Dr. Malone describes his prestigious academic training at the beginning, he describes a shift to industry, because he realized that “the world is full of these academic thought leaders that publish in big journals... I really wanted to make products that would help people,” and that since then he’s been “focused on actually doing stuff” (“Malone” [00:06:41 – 00:07:09]). This appeals to the stereotype that academic scientists, in their ivory towers isolated from the needs of real people, are unable to apply their research in real-world ways. Since

populism centers on the people vs the elite, this recasts academia as the knowledge elites, who are unresponsive to the needs of the common man because of their isolation from reality. This contradiction helps us better clarify that distance as a transcendent positioning, which retains the good while discarding the bad. To be precise, Goldenberg's conception of the maverick as appealing in "his distance from organized science" (162), is too conceptually simple for the alternative-expert. The vagueness of the word "distance" could imply both someone who left academia or someone who never obtained an education at all. Thus, we must go beyond the figure of the maverick to understand the appeal of the alternative-expert. Dr. Malone's language of having "left" academia demonstrates that he is in the former category, as someone whose educational and research credentials are not lessened when they leave the institutions surrounding academic research, and who sees the flaws of academia and claims leaving will allow him to do more. He simultaneously occupies the role of someone who "actually" does things and discards the negative affiliation with academia characterized as an ivory tower, yet simultaneously maintains the expertise implied by these academic credentials. We begin to see how the maverick has one foot in each camp: even after breaking away, they still retain the status gained from previous experience with academic institutions.

This narrative of institutional exodus is complemented by populist origin stories: As Dr. Malone says to Rogan in his intro, he was "originally a carpenter and a farmhand," and that "it was really bold to think that [he] could get into medical school" ("Malone" [00:01:30 - 00:08:55]). This rhetoric centers his role as an everyday figure and a man of the people, yet pairing it with his accomplishment of getting into med school retains the prestige and value of these credentials. Thus, the narratives transforms once again: the figure who leaves academia is just leaving academia, but the figure who starts as a common man of the people, has a temporary

climb up the ranks of academia, then decides to leave invokes the appeal of the prodigal son who has returned after his errance— yet this prodigal son still retains the benefits of academia accrued in his temporary absence from his true, populist roots.

Alongside narratives of an institutional exodus, the alternative-experts construct two categories of physicians: the academic, by-the-book providers, and the brave dissenting physician, willing to take risks based on what they know is correct. This further complicates outsider status, making them outsiders in their deviation from mainstream science but simultaneously, their characterization as a brave dissenting physician appeals to the cultural imaginary of what a physician should be, which replicates insider status. In this way, transcendent distance is again relevant— it is not just distance from organized science, but distance stemming from the fact that they actually reproduce inside status better than the organized scientists. In a discussion about monoclonal antibodies, Dr. McCullough criticizes doctors for not deviating from guidelines which limit distribution of monoclonal antibodies, criticizing them for being too weak to stand up to institutions for the good of patients. Monoclonal antibodies are widely considered safe and effective, even by mainstream experts, so the point of contestation that Dr. McCullough focuses on is the “arbitrariness” of their distribution: he says, “not only have they made it difficult to get In Texas, they actually put these parameters on who gets it and who doesn't” and supports this with a few anecdotes, including the following:

...you have to be in a high-risk ethnicity to get it. A friend of mine went, he had COVID, and he is a healthy Caucasian male in his 30s. And they told him, you are not qualified to receive the monoclonal antibodies. And the lady who's working there said, if you were another ethnicity, like if you were Hispanic or black, then we would qualify you. And she was like, look, this is not my idea. This is, I just have to follow the rules (“McCullough” [02:08:10 - 02:09:10]).

This quote creates a perception of arbitrary rules that are simply being followed by the lower level healthcare workers tasked with carrying them out. His rationalization for why doctors are not deviating paints them in a negative light, as simple rule followers because of their academic affiliation: “doctors are not like police officers or firemen or world wrestling champions. Doctors are kind of nerds. There's no check box that says, I'm courageous. I'm willing to take some risks... And I think there was a small number of doctors” who are willing to take those risks, pointedly sorting himself into the latter category of doctors by saying “I guess I'm one of them” (“McCullough” [02:13:00 – 2:16:00]) He creates a contrast between the doctors with courage, willing to act, and those who are too paralyzed to act; The latter are stalled into inaction because they are “nerds,” an academic word which further invokes the ivory tower stereotype of academics being unable to intervene in the real world. The former, on the other hand, are “courageous” like “police,” “firefighters” and “world-wrestling champions,” allowing him to maintain the credibility of a doctor while constructing a small class of doctors with courage to break the rules and take the risks that will help them save lives, creating their brave status.¹⁷ In this way, the alternative-experts create a tempered distance that allows him to maintain the credentials of a physician by rhetorically constructing two categories of physicians.

The alternative experts position themselves as part of the group of courageous doctors, increasing their expert credibility. Dr. McCullough supports his aforementioned assertion, “I

¹⁷ Though firefighters and policemen are traditionally viewed as emblems of courage, it is worth it to take a moment to explain the odd man out: the world-wrestling champion. Though the invocations of the other two figures are sufficiently self-explanatory to make my point, World-Wrestlers will be particularly well received by Rogan's predominantly male audience (Stecula 6) because they invoke the values of sportsmanship which journalism and communication Scholar Lascity argues makes up an integral part of Joe Rogan's specific brand of masculinity (Lascity 10). Moreover, Rogan first gained popularity for his MMA commentary (and his own role as an amateur MMA fighter) and listeners will appreciate his invocations of wrestling as a basis for MMA (Rogan, “#147 Gable Steveson”). Though this thesis is not interested in intent, but rather the rhetorical salience of strategies which, invoked either willingly or unwillingly, construct expert status in the mind of the layperson, it is worth noting that this oddly specific invocation implies an awareness on the part of Dr. McCullough of who his audience is.

guess I'm one of the [courageous doctors]" by positioning himself with this group as someone who fights arbitrary rules about monoclonal antibodies and takes risks for the good of his patients. He discusses sending his patients to get monoclonal antibodies at hospitals: "I say, get monoclonal antibodies in the ER... I was relentless... I was relentless...." ("McCullough" [02:13:04 - 02:15:58]). Specifically the wording "relentless" implies perseverance and willingness to stick to what he knows is right despite mainstream pushback. In this way, two categories of physicians are constructed: the nerdy, academic rule followers, who fail to adequately fail for their patients because they fail to take risks, and the populist, risk takers who have what it takes to truly care for their patients.

This rule-breaking ethos is populist not only because the rules and guidelines are established by the elite, but also because it represents skepticism toward established institutional narratives. Willingness to doubt the guidelines is a skepticism against the elite institution, making this a broader part of the populist appeal.¹⁸ This free-thinking, anti-institution populist skepticism also aligns with the cultural imaginaries of physicians as pioneering discoverers, demonstrating how this populist recasting of expertise relates back to the insider status. For example, Dr. Malone states that he "was the first to discover Famotidine," aka Pepcid, as a drug to treat COVID, because he became very sick with COVID and thought he was "going to die," so out of desperation, he took famotidine and discovered its efficacy ("Malone" [00:22:10 – 00:22:33]). This narrative of risk and self-experimentation likens Dr. Malone to scientific legends like Dr. Marshall, who struggled to convince the establishment that *H. pylori* bacteria caused ulcers, so he decided to swallow it himself to prove the link (Marshall and Adams).

¹⁸ Of course, this argument falls apart if the guidelines are actually correct. The mainstream experts are acting based on their assessment of the guidelines as correct. However, centering the debate around whether they are correct or not more rhetorically appealing than appeals to these populist, free thinking narratives.

Though Marshall's actions were eventually vindicated, he typifies the cultural imaginary of the lone scientist willing to take personal risks in defiance of institutional skepticism. The cultural imaginary of bravery and discovery takes on new meaning in our current context, aligning with the free-thinking ideals of populism. This speaks to how this narrative of bravery also relates back to inside status: the idealized biomedical expert takes risks, and recasting this brave expert under a populist light further pushes the academic expert away from the mainstream, while increasing the status of the alternative-expert. This rhetorical move pairs distance from the academic, elite establishment with admirable populist character traits about free-thinking skepticism towards the elites. This also recasts the cultural imaginaries of science under the appeals of populism, which overall, supports the individual alternative-expert's character and thus make them a more trustworthy alternative-expert overall.

Not only do the alternative-experts position themselves as having transcended from the academic elite, but returning to the political origins of the maverick, they transcend their government associations as well. COVID is a scientific issue, but with far reaching public health implications that necessitated extensive government involvement, much of which was intensely unpopular within the vaccine-hesitant community. In this context, it is not only the traditional boundaries of science– academia– that must be transcended in a recasting of expertise, but also governmental associations. As chapter 1 discusses, Dr. Malone's expertise is validated by his involvement with the CIA and DTRA. While he gains a boost to perceived credibility thus expertise through this involvement, he carefully situates himself as a critic of these fields, despite his involvement. Discussing his government positions, he states “this is not the branch that funded the Wuhan labs that's another branch of DTRA...” specifically distancing himself from positions that are heavily criticized while maintaining the status at DTRA (“Malone” [00:08:22 –

00:08:26]). He continues: “I’ve got many friends in the intelligence community, so I’m... kind of a pretty deep insider in terms of the government, I know Tony Fauci personally I’ve dealt with him my whole career...” (“Malone” [00:08:28 – 00:08:48]) It is a rhetorical act which claims access through insider status, literally through the word “insider,” and thus implies profound, intimate knowledge and access to the institution, while still implying a distance as he is willing to dissent from it. but also signals a willingness to disclose this knowledge or use it in ways that it was not intended to be used. Similarly, he discusses interacting with Dr. Antony Fauci, MD, who previously directed the National Institutes of Allergy and Infectious Disease and served as the Chief Medical Advisor to President Biden during 2021, across his entire career, but instead of saying he “worked” with him, as would be typical for collegial colleagues, he “dealt” with him, implying a distaste or other negative opinion. This careful positioning with proximity to government institutions, but not in an uncritical way, straddles between insider and outsider status, allowing them to claim the benefits of both camps, and discard the critiques of either camp. This critical positioning maintains a distance from the institutions, which are seen as corrupt and thus untrustworthy, while still allowing them to maintain their credibility. Each independently boosts rhetorical expert appeal, and maintaining this transcendent distance maximizes the expert status.

Establishment vs Alternative-Expert

A rebel without a cause appeals to no man; thus, the maverick rhetorically succeeds only when he has an enemy. Though I argue against a simple understanding of populist outsider status as only opposing the institution, it is still a salient rhetorical move. Exploring it further reveals how they not only attack from a place of populism, but once again invoke insider norms, recast through populism, to assert why they are a better option than the mainstream.

The Corrupt Institution vs Values of Health and Healing

"...Unfortunately, Ivermectin has... affected the most sensitive organ on humans, the wallet"

- Dr. Kory ("Weinstein & Kory" [01:46:51 - 01:47:00]).

Indeed, the maverick's *raison d'être* must manifest as an establishment to oppose. They impugn broader mainstream institutions as entangled in corruption by the profit-motives of the pharmaceutical industry. This sows mistrust, as science and the establishment not only need to be epistemically trustworthy, but have good intentions, to be viewed as credible. These intentions stem from the cultural imaginaries of biomedicine's intentions, which as just as important as their epistemological strength in creating trust: As Goldenberg writes, it is not just the factual basis of scientific findings, but "the goodwill of scientific enterprise—to improve lives and promote public interests—[which] needs to be trusted as well" (Goldenberg, 141). Indeed, as Starr writes on the basis of physician authority, the profession of medicine's "power is avowedly enlisted solely in the interests of health... On this basis, physicians exercise authority over patients, their fellow workers in health care, and even the public at large..." (Starr, 5-6) In other words, the trust in experts is undercut if viewers perceive that motives other than the health of others are driving their goals. Moreover, the internalization of these values by the

alternative-expert physicians functions as a form of embodiment capital, demonstrating how an insider appeal continues to be salient even when creating distance from the establishment and seemingly invoking outsider status. Though Starr's claim is rooted in physician professional authority, it can be extrapolated to professions— such as scientists— more broadly, especially because as Goldenberg demonstrates, scientists similarly stake their credibility on their commitment to the knowledge-making practices of their field. This demonstrates the overall role of an orientation towards healing as a form of embodiment capital that constitutes insider expertise.

Within populist meta-narratives, the machinations of the elite are characterized as driven by material gain. Political Scientist Sarah Engler identifies the overlap between anti-corruption and anti-elite rhetoric in populism in her article, “Fighting corruption” or “fighting the corrupt elite”? Politicizing corruption within and beyond the populist divide” (Engler). Profit figures prominently as an obvious corrupter. Indeed, by reframing traditional scientific, governmental and other large institutions as entangled together in corrupt, profit seeking motives of the pharmaceutical industry aka *Big Pharma*, a colloquial term for the world's largest, most influential multinational pharmaceutical companies, these institutions are cast as the elite in populist metanarratives. As the profit-driven elite, trust in these mainstream institutions is undercut. The alternative-experts deploy the logic of profit as a systemic indictment: it is so vast and structurally embedded that it pulls every adjacent institution from regulatory agencies, academia, and media— into a corrupt, elitist orbit. This systemic corruption delegitimizes the mainstream, creating a void of expertise and leadership for the alternative-expert to fill. By contrasting the elite's profit-driven motives with their own value orientation towards health— which, as Starr and Goldenberg both establish, is a constitutive feature of biomedical authority—

the alternative-experts do not merely attack the mainstream with a populist position, but actively reproduce the insider values that underpin the cultural imaginaries of biomedical expertise.

Part A: Entangled in Profit– “Big Pharma” and The Malicious Elite

The alternative-experts paint pharmaceutical companies, regulatory agencies, and media as entangled in profit-driven corruption that prioritizes financial gain over the health of the American people, which delegitimizes mainstream expert recommendations under populist metanarratives. The first step in this rhetorical move is illustrating the corrupt, profit driven motivations and thus the elite character of Big-Pharma with populist rhetoric. Dr. McCullough, for example, characterizes vaccines as intensely profitable, both because the US government purchases them and pays for them to be developed. He states, “Do you know what, the vaccines that Pfizer [made], in his first year, hit 33 billion? And now I think next year, [it’s likely they will turn a profit of] 36 billion, [with] no development costs,” because the “government paid for the development costs” (“McCullough”, [02:39:23 – 02:41:08]). His claims paint Pfizer as a profit-hungry corporation that makes billions of dollars off of tax-payer’s money, successfully casting it as an elite institution against the people in a populist metanarrative.

Similarly, Dr. Kory reinforces the narrative of profit over science by contrasting the cost of Remdesivir–“\$3,000... [per] dose drug?”–with ivermectin, saying “there’s no money to be made off ivermectin” because it has been manufactured as a generic for so long (“Kory & Weinstein” [00:08:16 – 00:08:25]). He goes on to imply that the broad prioritization of drugs like Remdesivir is due to the profit motivation for pharmaceuticals. He claims, “there are studies showing some support [for Remdesivir]...But its impact is actually minimal. And [the] studies vary... WHO does not recommend [Remdesivir]. And they did a big trial of Remdesivir. They showed that it did not help anyone, and so they don't recommend it” (8:34-9:02). His invocation

of scientific studies and the official recommendation of the WHO, a reputable and massive global health organization, lends credence to his claims. The choice to use Remdesivir over ivermectin, then, appears to be a profit-driven decision rather than a scientific one. The alternative-experts characterize this position as especially egregious by purporting that profit also drives both the search for new drugs in lieu of using ivermectin and the use of the COVID-vaccines. As Dr. Weinstein states:

“Anthony Fauci gave a press conference about a \$3 billion initiative to find drugs that work against COVID. Now of course, these drugs that they find will all be under patent and therefore highly profitable. So what you’ve got... is drug companies, [such as] Merck [which] is involved in Molnupiravir, this new drug. It is also involved in an agreement with Johnson & Johnson to distribute their vaccine, and strangely, we are ignoring the evidence that is right in front of us that we have multiple drugs that are highly effective for COVID and one, that I would point out again, is highly effective as a prophylactic... shareholder value must be driving things behind the scenes” (“Weinstein & Kory” [01:14:42 - 01:15:35]).

His statement suggests that the government is subsidizing the discovery of new drugs and the vaccine, so pharmaceutical companies like Merck and J&J are financially incentivized to downplay the efficacy of other drugs. This sows doubt in these product’s efficacy while providing a reason for the mainstream to be financially motivated to rebut their claims about the science and effectiveness of alternative-drugs like Ivermectin. Their accusations set up a divide between the mainstream experts who promote specific interventions for profit and themselves, maverick outsiders who are unafraid to challenge corrupt institutions and present the true science to the people.

Dr. Malone, too argues for a similar corruption of big-pharma as corrupt by profit, extending the condemnation to their amoral character. He states that pharmaceutical industries are “not grounded in the ethical principles that you and I, as average people, believe in. They don't live in that world..they are about profit- return on investment... the overlords that own

them BlackRock, Vanguard, State Street, etc. are completely decoupled from nation states, have no moral core- they have no moral purpose. Their only purpose is return on investment” (“Malone” [02:36:10 – 02:36:30]). This explicit condemnation of profit seeking behavior emphasizes the role Big-Pharma plays as an elite, corrupt force in populist metanarratives. Dr. Malone further invokes populist metanarratives centering the individual by appealing to “you and I” as “average people” and the ethical principles which he purports guide these average, everyday individuals. Moreover, the invocation of pharma as “decoupled” from “nation states” implicitly sets up a contrast between the two: pharma, painted as profit driven and corrupt, functions differently from the democratic mandate of (the American) nation-state. The most basic moral purpose of a nation state: governance for and by the people, invokes the democratic purpose of nation-states. Appealing to this broader rhetoric of democracy highlights the intertwinement of democratic and populist appeals as rooted in serving the people, and reveals the broader rhetorical utility of these claims in a democratic nation.

Alongside condemning pharma, the alternative-experts extend their indictment of corrupt, profit-driven behavior to mainstream physicians, implicating doctors as willing participants in the corrupt network. This offers a justification for the broader condemnation of mainstream experts, not just of Big-Pharma, weaving together mainstream institutions as prioritizing financial gain over the health of the American people, which delegitimizes mainstream expert recommendations. Dr. Malone, for instance, states, “Pfizer [a COVID vaccine manufacturer], is one of the most criminal pharmaceutical organizations in the world based on their past legal history and fines. What do those fines include? Bribing physicians” ([02:35:55 – 02:36:08]). The significance of this accusation is structural: as established in Chapter 1, the physician's claim to authority rests not only on epistemic competence but on a moral orientation toward the health of

others. Acting for profit, not for health, violates the insider value orientation that constitutes expertise. Connecting the criminality of an already mistrusted entity with the mainstream physicians who prescribe their treatments essentially dissolves the boundary between pharma and mainstream physicians, casting both as corrupt, profit-seeking elites.

Like physicians, the mainstream US government's public health system is similarly portrayed as influenced by the profit-procuring mechanisms of pharmaceutical corporations. Dr. Kory does this by connecting pharmaceutical profits with US health regulatory agencies, stating, "we live in a public health system, which favors for-profit medicines over non-profit... the things that are favored are those with financial interests, and so that's who gets the ear of the agencies, that's who gets attention by the FDA" ("Wesintein & Kory," [01:18:05 - 01:18:54]). Pointing out a salient, common critique of pharmaceutical development— that inexpensive and unprofitable drugs are rarely developed due to profit concerns, and then connecting this critique to the FDA's review of drugs, condemns the FDA. Rogan responds in agreement, "the influence of the pharmaceutical companies is a real thing. It's global," demonstrating how the claim that pharma exerts an undue influence on the public health system resonates ([01:19:09 – 01:19:13]).¹⁹

This rhetoric of profit driven corruption implicates an extensive network of mainstream institutions, condemning them all as elites sacrificing the health of the American people in the name of self-interest in populist metanarratives. From Rogan, we learn that the claim to persuasive influence of pharmaceutical companies is quite convincing: This is demonstrated by

¹⁹ The media is briefly implicated in this network as well: Dr. Malone discusses the Trusted news initiative, which purportedly limits misinformation but which he claims "makes it really easy for the media to..." remove content/ deplatform "...anybody that's saying something that is contrary to the interests of... the vaccine industry" (Malone [00:46:00 – 00:48:00]). In this way, the media is positioned as directly deferring to the profit interests of the pharmaceutical industries, which ties them up with the corrupt elite.

Rogan's repetition of the persuasive influence of pharmaceutical companies that the alternative-experts claim. He discusses:

“how people have just decided because they're scared and because they want a solution that the pharmaceutical companies have their best interests at heart and that they're not these machines that are designed to make money.... the drugs are often beneficial, but their main goal is to make money and if they can fudge the data if they can move the numbers around if they can delete negative consequences [they will]” (“Malone” [02:35:23 – 02:35:56]).

The repeated “ifs” seed doubt in the intentions of the pharmaceutical industry. His own engagement with this rhetoric discusses the appeal of this narrative of the pharmaceutical industry as selfish, profit-driven and corrupt and entangling the mainstream in their corruption— a move which is rhetorically salient to the layperson.

These populist claims to corruption, specifically of Big-Pharma, have a very widespread appeal. Part of this stems from their interwoven nature with the “for the people” orientation of democracy, which is an important ideal in American culture. While this specific claim about COVID corruption is fictitious, the general condemnation of big pharma and the web of corruption is incredibly appealing because it is factual in other instances. Dr. Malone's claims that Pfizer bribed doctors are not incorrect— they had to pay 2.3 billion in 2009 for promoting their drugs to doctors for medically unapproved circumstances, and violating anti-kickback statutes by flying them out to resorts and providing them with high-end gifts (“Pfizer Pays Record Settlement in Marketing-Violation Case”). Similarly, Sergio Sismondo's book *Ghost-Managed Medicine: Big Pharma's Invisible Hands* (2018) explores how pharma systematically shapes the production and dissemination of medical knowledge from the inside— funding studies, ghostwriting journal articles, managing publication strategy, and placing industry-favorable research in reputable peer-reviewed journals, all while maintaining the appearance of independent science. The result is that what looks like objective medical knowledge is actually

deeply intertwined with corporations and their profit concerns. A similar concern emerges with policy: physician Marcia Angell argues in *The Truth About Drug Companies* (2004), the immense profit of the pharmaceutical industry allows them to push their agenda through Congress, the FDA, and academic medical centers. Together, this creates a damning picture of the intertwining of Big-pharma and other large mainstream institutions, allowing them to be easily implicated by populist rhetoric, even when no fault is present.

Part B: The Alternative Expert as Uncorrupted by Profit

In light of the corruption of the institution, the perception of unselfish value orientations towards the health and wellbeing of others create expert status for the alternative-experts. Notably, the values of health and moral character should signify insider status, as they function similarly to the necessity of academic credentialing and consensus: as Starr and Goldenberg both write, these values constitute physician and scientist authority. The way the alternative-experts invoke and perform these values functions as embodiment capital, allowing them to reproduce insider status even within an ostensibly outsider rhetorical position. This complicates any simple characterization of their expertise as purely populist or outsider.

However, it is not only insider expertise: This value is recast in populist frameworks, complicating the purely insider status. As Meyer concludes in his reconceptualization of expertise under populism, populism 'allow[s] expertise to be distinguished from elitism and to be recognized in caring relations to the subjects of knowledge' (Meyer 845), suggesting that a genuine orientation towards health does not solely reproduce insider status but takes on new meaning within a populist context—effectively being recast into a new form of expertise.

This value orientation towards health is frequently invoked by the alternative-experts, creating insider status and expertise which complicates the outsider appeal. Indeed, Dr. Kory states that “There's- there's no one championing Ivermectin, except for like my little group of, uh, non-profit doctors who became expert at Ivermectin” invoking the role of their non-profit to show that their advocacy is due to scientific rationale, not monetary gain (“Weinstein & Kory” [01:17:43 – 01:19:09]). He similarly states, “we don't make any money off of [our support of Ivermectin], we're just trying to doctor” ([00:38:09 – 00:39:50]), furthering the idea that their motivations are genuinely oriented towards health, which as Starr points out, strengthens their expertise. He extends this moral position to the broader constructed community that studies ivermectin by pointing out that there is currently a large trial of ivermectin funded by “philanthropy” instead of by big-pharma. Philanthropic contributions are often associated with morally upstanding causes, which paints ivermectin in a more favorable light than the government and big-pharma funded vaccines. By doing this, he appeals to communal validation (as discussed in chapter 1) and constructs a community of alternative-experts who are doing this research without the profit orientation in mind, furthering the moral position of these alternative experts. In fact, not only do they advocate for these drugs with no monetary gain, they position themselves as willing to do it despite the monetary risks and other damage which it inflicts upon them. In Dr. Malone’s podcast episode, he brings up Dr. McCullough as an example of the monetary cost of dissent, stating: “Peter McCullough is the textbook example of hunting physicians, right the guy is... 150,000 in debt right now in the hole in trying to defend his medical license,” demonstrating how the alternative-experts frame their efforts as not only unprofitable but as actively financially damaging (“Malone” [01:14:53 – 01:15:02]). Whereas mainstream products such as vaccines and Remdesivir are framed as profit-driven interventions,

they detach ivermectin and their work from profits, purifying their reasons and appearing to be value driven by health orientations. Due to the reliance on a scientist and physician's value orientation on their trustworthiness and expertise, this rhetoric improves their status as experts.

The alternative-experts cement this point by explicitly stating their values and goals are aimed at helping the American people and improving health. As Dr. Kory states, they are doing this because "we took an oath as physicians to help patients. When this thing came to our shores, all we wanted to do is learn as much as we could about this disease to figure out how to kick its ass" ("Weinstein & Kory" [00:38:09 – 00:39:50]). The invocation of the doctor's Hippocratic "oath" to help invoke the symbolic capital of value orientation towards patients which legitimizes physicians, and Dr. Kory's status as a doctor and expert is heightened because of it. On the other hand, the alternative-experts characterize the mainstream-experts as not fulfilling this responsibility towards patients because of a proposed selfishness which undercuts their expert character: Dr. McCullough claims,

Doctors for the first time in their lives felt like they could get the disease themselves... All the discussion was on personal protective equipment, hand sanitizer [and] negative airflow rooms. It was all about protecting the healthcare workers. There wasn't any focus on sick patients ("McCullough" [03:45 - 04:19]).

This both undercuts the mainstream as timid and selfish, while making the alternative-expert's commitment to helping patients stand out in comparison. Continuing this theme, Dr. Malone states they are doing this not because there is "some financial incentive" but solely because "it's the right thing to do" ([01:25:00 – 01:25:26]). Dr. Weinstein similarly states,

[W]hat is motivating us is that there is a lot of- there's a lot of risk to human beings out there. Just even the loss of one person so devastates a family that just thinking about all the people who are gonna be harmed by the fact that we're not following the evidence and figuring where it leads, it's just-- There really isn't a choice but to talk about it, ("Weinstein & Kory" [02:50:48- 02:51:15])

Zooming in on the meaning of a single human life and the impact on their family is humanizing, and makes Dr. Weinstein's character stand out as admirable. His focus on "following the evidence" similarly functions as symbolic capital of science, as science as a knowledge making system and the rational application of it is highly valued. Indeed, Rogan's response to this rhetoric demonstrates its efficacy amongst his viewership. Dr. Weinstein states, "I think what-what keeps stopping me and my tracks is the magnitude... The amount of needless human suffering [from not utilizing Ivermectin] is almost incalculable," to which Rogan repeats the final word back, "incalculable," in a tone of disbelief, affirming Dr. Weinstein and marking the horror of the purported tragedy of not using ivermectin ("Weinstein & Kory" [01:19:19-01:19:37]).

While mainstream-experts would argue that they pursue mainstream treatment paths for the same values of health and for the good of patients, they do not rhetorically state it repeatedly in the way the alternative experts do. Dr. Sanjay Gupta did not spend any time discussing why he was dedicated to broadcasting his claims about COVID and COVID vaccines as a science communicator, instead opting to rely more on the science itself in the podcast episode. It is highly likely that Gupta and other mainstream experts are engaging in this advocacy for the same reasons as the alternative experts claim to be doing so: for the health and good of the American people and the world. However, combined with the condemnation of the mainstream as either seeking profits of, the alternative experts maintain more credibility in this context.

While the alternative-experts steer clear of profit and their characters are morally appealing, it is important to note that vaccine hesitancy generates a lot of revenue— Though it is impossible to truly know what drives Dr. Malone, Dr. Kory, Dr. Weinstein and Dr. McCullough's claims, and it is fully possible that their advocacy is genuine, it is important to understand the

financial context in which they operate. For example, Dr. Malone founded a company which is a direct competitor to the RNA vaccines, giving him a profit motive to denigrate the current COVID vaccines. Yet, Dr. Malone only cursorily mentions that he is the founder of “Inovio,” stating “they have one of the DNA vaccines for Covid” before quickly transitioning away (Malone [00:01:30 - 00:08:55]). His involvement with this company creates a conflict of interest, and his claims must be scrutinized in the light of his financial stake.

Moreover, these figures all generate revenue from posting vaccine-hesitant claims on their respective social media platforms. For example, in Rogan’s podcast with Dr. Weinstein and Dr. Kory, we learn about the financial motivation when Rogan starts in-media res, stating “I’ve never had to do an emergency podcast before...” (“Weinstein & Kory” [00:00:15 – 00:00:19]). He then gives the guests time to introduce their credentials, in which Dr. Weinstein introduces himself as the co-founder and creator of the Dark Horse Podcast, a podcast where and his wife use their scientific backgrounds to follow the “developing COVID story” ([00:01:21 – 00:01:31]). Rogan later clarifies why this emergency meeting has been called- not because of the emergent situation of COVID and to disseminate information to his viewers, but instead, “the reason why we’re here is [Dr. Weinstein’s] channel is in jeopardy,” as it has recently been taken off of youtube ([00:09:49 – 00:09:53]). Dr. Weinstein directly states that “a majority of my family’s income comes through our two YouTube channels,” demonstrating a clear financial motivation for his channel to continue to stay up and flourish ([00:10:19 – 00:10:27]). The sensationalism of his claims has built a liveable salary for his entire family, and he has a vested interest in continuing to draw eyes towards him.

Vaccine-hesitant content can be extremely profitable: In 2022, the Center for Countering Digital Hate published findings concluding that five substackers draw in a combined \$2.5 million

a year from their vaccine-hesitant centered blogs. The top two substack earners, Joseph Mercola and Alex Berensen, collectively account for 2.2 million of those earnings, but one of the others in the top five is actually Dr. Malone, leaving him a chunk of a hefty sum, 300,000 dollars per annum. Dr. Kory's "little-group of, uh, non-profit doctors," or the FLCCC,²⁰ (Weinstein & Kory [01:17:43 – 01:19:09]), brought in \$1 million in 2021. The American Frontline Doctors, one of the numerous organizations Dr. Malone states he is involved with²¹ in chapter 1, brought in \$21 million in 2022. According to publicly-available salary records, Dr. Kory was paid \$368,815 as FLCCC chairman in 2022 (Weber). This is only what is publicly available, and likely only reveals the tip of the iceberg in the profits made from ventures that directly depend on vaccine-hesitant sentiment. To be charitable, I will say that maybe they were simply incorrect about their claims. Regardless, the actual profits they make undercut the strict dichotomy they setup between themselves as in it for the good of the people and the mainstream as doing it for profit, their rhetoric in the podcast episodes is convincing, and these financial motive does not undercut their credibility of expertise amongst JRE listeners because they are barely known.

The Ignorant Masses vs Competent Alternative Experts

The previous sections explore the role of profit-driven corruption in the pharmaceutical industry and as entangling broader mainstream-institutions. Overall, these rhetorical strategies work well because they are deployed within a context where populist metanarratives about the corrupt elite conspiring against the people are common. However, this rhetoric is inherently

²⁰ Which Dr. Malone is also a part of.

²¹ Alongside the FLCCC: As previously discussed in chapter 1, Dr. McCullough states "Many of us are members of the Association of American Physician and Surgeons, or the Frontline Critical Care Consortium, or American Frontline Doctors, or the Truth for Health Foundation" (McCullough, 02:23:56 - 02:26:33).

based around assumptions of malicious, selfish intent. Claiming widescale malice runs the risk of becoming implausible thereby undercutting the credibility of the alternative experts' accusations. Despite some ostentatiously conspiratorial claims, the alternative experts hedge accusations of malice with invocations of ignorance and fear as explanations for mainstream acceptance of COVID vaccines. This rebuts the objections and critiques of conspiratorial thinking they face. To explain the wider acceptance of mainstream claims around COVID and vaccine-science, they invoke the ignorance of the masses of laypersons and even some physicians in uncritically internalizing or reproducing the narratives of the corrupt elite. They paint this uncritical engagement as stemming from rational fears– the pandemic posed a threat to all, and left people scrambling for answers. They construct a rhetoric of ignorance which drives what they would call an erroneous acceptance of mainstream-solutions to the pandemic, such as vaccines. For example, Dr. Kory offers an analogy which underscores this point about a corrupt elite vs those who are being led astray, stating, “ a plane is crashing...and everyone's saying, ‘Listen to the captain,’ ... And no one's considering, what if the hijackers already got the captain?” (“Weinstein and Kory” [01:55:37 – 01:55:51]) This analogy maps neatly onto the populist narrative the alternative-experts construct: the hijackers represent the corrupt elite who have captured mainstream institutions, the passengers represent the fearful, misled layperson. The fear of the COVID 19 situation has allowed the laypersons to be led astray.

To be clear, this rhetoric does inherently imply malice from the part of a select, elite few. But since that claim maps neatly onto populist metanarratives of how the world works, it is rhetorically appealing. Moreover, it allows the alternative-experts to emerge as heroic comparatively,

there is an organized opposition who are now understanding that some of these agencies are captured, and that if they keep listening to them, they're gonna keep getting what's happening, which is uncontrollable spread, uh, crisis situations...India, finally, they broke free from the WHO....[one place in India called] Uttar Pradesh... [has] some of the best numbers, uh, not only in India and in the world.

Along with fear, Dr. McCullough frequently cites ignorance as a driving factor of acceptance to mainstream solutions, further diversifying rhetorical claims as to why the mainstream is erroneous and thus strengthening his argument and expertise. For example, Dr. McCullough discusses hydroxychloroquine, asserting that while it was in fact suppressed: “I saw a trend starting in April, May and June, where it became clear that anything we were doing to try to help patients with early treatment was receiving a chill. And the chill was coming through academic institutions through the medical literature” (McCullough, [00:25:36 – 00:29:02]). While he frames himself as attempting to “help patients” as an advocate of early treatment, condemning the institutions, he does not ascribe these actions to malicious intent. In fact, he says that there was a moment when people began to recognize that the spike protein on the virus could be neutralized through “vaccine-induced antibodies:” he continues,

Once that became abundantly clear in the literature, there appeared to be almost a lockstep developed, where people said, aha, that's it. That's the solution. We're going to vaccinate our way out of this problem. We don't even need to worry about how to treat the problem. We don't need to hear about drugs to treat the problem. and the enthusiasm and the hubris for vaccination spread across academic medical centers all over the country” ([00:29:59 – 00:30:45]).

In other words, they thought they had found a solution, so they simply stopped focusing on any other solutions. People are likely unwilling to believe that almost every doctor is simply willing to jeopardize the health of the American people and the world for profit or other

malicious motivations. This rhetoric of ignorance circumvents this conspiratorial problem, by attributing widespread vaccine acceptance to groupthink dynamics, the alternative experts claims become more plausible.

This is demonstrated by Joe Rogan’s question to Dr. McCullough, “how do you get so many doctors to go along with this?... to not speak out against the lack of pre-hospitalization care, the lack of early treatment?” which demonstrates the incredulity that all doctors could simply fail to “speak out” against the suppression of early treatments such as hydroxychloroquine, a substance which has been already established as scientifically and ethically important in terms of treatment (McCullough [00:39:00 – 00:39:13]). Dr. McCullough replies that he believes there are only about “500 doctors who knows what's going on,” implying it is not an act of malice on the doctors part, but rather that they simply are ignorant of “what’s going on” ([00:39:00 – 00:41:28]). He affirms this by saying that the “nurses are more awake,” cementing this idea that there are those who are ignorant and those who are not ([00:39:00 – 00:41:28]). The word “awake” is particularly revealing in the context of the manosphere viewers of the podcast, as it alludes to a unifying philosophy of redpilling which will be rhetorically well received.

Redpilling is a philosophy borrowed from the Matrix, in which the protagonist, Neo, recognizes there is something odd about reality, and is offered a choice to either take the bluepill, which will allow him to remain in in his deluded, comfortable simulation, or to take the redpill, which symbolizes choosing to face the ugly truth of reality and literally wake up from his containment pod. In her article “Alphas, Betas, and Incels: Theorizing the Masculinities of the Manosphere,” Professor of Digital Media and Gender Debbie Ging, in a study of the Involuntary celibate community r/TheRedPill or r/TRP for short, writes that “the Red Pill philosophy

purports to awaken men to feminism's misandry and brainwashing, and is the key concept that unites all of these communities" (Ging 3). Though she applies it to manosphere communities in r/TRP, this concept is seen to function within the context of this podcast episode, but instead of feminist brainwashing, it is pharmaceutical brainwashing that one must "awake[n]" from.

Dr. McCullough continues to expand the scale on which this brainwashing functions, cementing the redpill philosophy in the context of these claims about COVID. He claims, it is "leaders of the major churches, every single one of them..." and "every major global international leader is under the spell" who are deluded under a "mass formation psychosis," which he defines as when "group think that develops that's so strong that at least to something horrific," using suicide cults and the holocaust as examples ("McCullough" [00:39:15 – 00:41:28]). Groupthink, while not particularly charitable, implies an ignorance which is different from direct malice, asserting that there are many reasons why so many people take up the vaccines despite their purported scientific failings. Indeed, the language of "spell" continues to imply a deluded, wool-over-the-eyes perspective, imposed upon one, which detracts from the malice involved in it. He cites two credentialed scholars with the "Dr." title, one of which is a "psychiatrist," to bolster his claims, appealing to the communal validation and academic qualification based institutional capital of chapter 1 ([39:15 - 41:28]). Malone repeated this claim in his own podcast episode, zooming back in on the example of Nazi Germany and stating that this is how they had a "very intelligent highly educated population... "who "went barking mad," demonstrating how people became incredibly misguided but decentering the malice from the vast majority of people ([02:36:30 – 02:38:23]). After Dr. Malone's episode, many fact-checkers and news articles specifically singled out the mass-formation psychosis claim as particularly unsubstantiated, including the 270 mainstream-experts who wrote an article to spotify, yet this

fact-checking is unlikely to make a dent when it aligns with a deeply held philosophies which unify these communities.

Indeed, the rhetoric of fear of the masses inadvertently allowing the corrupt elite's machinations to come to fruition is shown to be rhetorically salient, as Rogan also engages with it. He states that people accept suspect pharmaceutical products like vaccines "because they're scared and want a solution," the alternative-experts invoke the general fear surrounding the COVID pandemic and the want to resolve it as an emotional fear which is clouding decision-making. Moreover, the word "people" is a vague phrase, and could reference laypersons, doctors, media or the government alike, granting a widespread boon that improves the rhetorical appeal: It is not all elite, corrupt bad-actors, but there are enough that others are swept up in their uncritical ignorance.

The Dogmatic Scientific Elite vs the Heretics

Moreover, by painting mainstream science as dogmatic and themselves as opposing heretics, the alternative-experts tap into one of the most powerful cultural imaginaries in the history of science: the lone dissenter who is proven right despite total institutional opposition. Indeed, Goldenberg writes that Wakefield capitalized on the role of dissent and hereticism in science's history to gain public admiration as a "David figure against the Goliath of institutionalized science" (153), establishing the rhetorical appeal of positioning oneself as a brave dissenter against a dogmatic establishment. This appeal takes on a significant outsider positionality in its opposition to mainstream science, yet simultaneously, their heretical narratives are rooted in the cultural imaginaries of science. Through narratives of hereticism recast through populist free-thinking, they construct a position that is oppositional enough to

appeal to anti-establishment sentiment while remaining insiders simultaneously. For this heretic rhetoric to function, however, it requires an establishment worth opposing. The next section outlines how the alternative-experts construct mainstream science as a dogmatic institution more akin to religious doctrine than rigorous inquiry, allowing room for their heretical narrative to construct a new expertise.

Part A: Dogmatic Elite

The alternative experts use the language of dogmatic religion to construct a vision of mainstream science as an edict rather than a scientific process. This creates a depiction of mainstream science as imposing narratives of science from the top-down, starkly contrasting the expectations that science is skeptical, derived from data, and unconnected to power dynamics. This theme is advanced explicitly, consider when Dr. Kory discusses the trusted news initiative, which is a “consortium of major media outlets “ who “suppress medical misinformation” (‘Weinstein and Kory” [01:55:14 - 01:55:23]). He states that based on their practices, medical information is “somehow defined as anything that doesn't come from what I call the gods of science and knowledge.... So from the leading agencies” ([01:55:23 - 01:55:31]). The explicit invocation of agencies operating as “gods” undercuts the idea that science is based around organized skepticism, not conformity to a pre-approved narrative. Moreover, it entangles the media in the efforts to advance these dogmatic scientific findings, creating a broader narrative of dogma.

Similarly, Dr. McCullough emphasizes the dogma of science by invoking a familiar talking point about the legitimacy of science:

McCullough: “Haven't you heard follow the science?”

Rogan: [giggles] “yeah.”

McCullough “Haven't you even heard that someone claims that they are science?”

Joe Rogan: “Yes I have” (“McCullough” [02:01:14 - 02:04:42]).

Someone who “is” science opposes what science is supposed to be, as it focuses knowledge making legitimacy in one person as opposed to the organized skepticism of the process. The cultural imaginary of science as skeptical underlies this claim, but instead of belying insider claims as it did in chapter 1, it allows the alternative-experts to tear down mainstream-institutions.

This rhetoric of the dogmatic mainstream calls into question the consensus. It also entangles the government with the dogma of mainstream science. Dr. Weinstein appeals to a notion that science is being inappropriately manipulated by the powers that be. Using the example of plate tectonics, Dr. Weinstein says,

[A] normal scientific consensus looks like, you know, plate tectonics, right? Plate tectonics was an absolutely heretical idea when it was introduced, the idea that the continents are moving, wow, that was mind-blowing and blowing and almost nobody got it at first, right? Today, everybody gets it... consensus arises from the work, from people challenging each other and discovering that, yeah, that thing seemed to make sense but it doesn't make sense when you look at the evidence (“Weinstein & Kory” [00:30:28 - 00:31:52]).

Though not all new consensuses are actually dramatic schisms with intense backlash, he paints scientific consensus as something that emerges slowly over time, and something that initially faces a lot of confusion and misunderstanding and in fact even “heretical,” implying new consensus must break old paradigms.²² This is not a necessary condition to consensus formation, but popular imagination of the history of science grants this mechanism of consensus formation, and thus the heretics, their appeal. However, his constructed idea of consensus, with an emphasis

²² To use Kuhn's language.

on doing the “work” and “challenging each other,” rightly appeals to the popular imagination and cultural imaginaries of how science works— especially the mertonian ideal of skepticism, as previously discussed. He claims the mainstream scientists, on the other hand, are not following this path to a “normal” scientific consensus in the case of COVID and COVID vaccines. He states, “These consensus... are being handed down from on high... It has no right to be labelled as scientific because it isn’t, it didn’t arise through the normal process. It isn’t what most people think it is— it is an official position...” (“Weinstein & Kory” [00:31:54 - 00:32:31]). This depiction of mainstream consensus as being “handed down” as opposed to rigorously researched, questioned and defended contrasts starkly with the ideal, “normal” consensus process that Dr. Weinstein depicts in the case of plate tectonics. This contrast, which makes visible the ideal, makes the mainstream appear even more corrupt in comparison²³. He also refers to the “officialdom” in the formation of these dogmatic consensus, Using the stem -dom likens government officials to a Kingdom, evoking imagery of edicts handed down on high and monarchical power concentration, which undercuts the democratic and populist values of the viewers. Meyer concurs, writing, “The first form of anti-populism is best captured in the slogan, “Follow the science,” demonstrating the ant-populists engage in the rhetoric that the alternative-experts push against. (Meyer, 2024, p. 848) Indeed, Those who challenge this authority, as Ranalli notes, “generally bristle at the label “denier,” claim that the professional scientific community is being inadequately skeptical and so the consensus that has been achieved is suspect,” speaking to the role of mainstream consensus being painted as unscientific (Ranalli 190). This problematization of consensus, alongside the rest of the mainstream science and

²³ Of course, this construction does not take into account the work back in the 90s Kariko was doing on the mRNA vaccines, or the mobilization of the entire world’s scientific workforce into solving this one problem, as an explanation for why this occurred so quickly.

institutions through dogmatic religious characterization, creates a damning depiction of the mainstream scientific processes.

Moreover, it heightens the alternative-experts position by giving them something to stand against. The alternative-experts utilize the rebellious language of the religious opposition to situate themselves as the maverick opposing this mainstream corruption of science. Indeed, the use of religious opposition within the framework of mainstream science operating akin to religious faith doctrines makes the alternative-experts more appealing, cementing their expertise among JRE viewers.

Part B: Heretic Experts

Similarly, their characters are constructed within terminology of religious opposition, which allows the alternative-experts to gain expert status left in the void of the dogmatic, discredited elite. As a claim both to the insider imaginaries of science and as a pushback against the establishment, this presents another dual positionality which establishes the elite's status. The mobilization of the language of the heretic is the prime vehicle for establishing this narrative: Early on in the episode, Rogan says to Dr. Malone, "the pandemic broke out" and now "you're kind of thought of as a heretic now in some strange way..." demonstrating Rogan's own recognition of the alternative experts through this lens ("Malone" [00:08:50 – 00:09:05]).²⁴ Dr. Weinstein's comments explain the appeal: he describes those who speak out against the mainstream narrative on ivermectin as "Heretics," and says the establishment is relying on the fact that people won't believe them on that basis: "Who do you believe? You're gonna believe the heretics?" ("Weinstein & Kory" [00:33:04 – 00:33:06]). However, he argues, heretics "actually

²⁴ Dr. Malone actually replies, "Pariah," broadening the excommunication from religion to society at large [00:08:50 – 00:09:05].

are an interesting group and the thing that unites them seems to be their independence of the structures that are controlling others" ([00:33:06 – 00:33:17]). Specifically, this construction of heretics as “independen[t]” of “control” is particularly appealing to the viewership of Rogan, whose brand revolves around being a free-thinker (Stecula). Moreover, the idea of the heretic goes back to the popular history of science, in which Galileo is mythologized as a sole scientist willing to speak out against the church’s geocentric dogma. This mythology has a potency in the cultural imaginaries of science, as Goldenberg writes, it functions as an important historical trope in science (153). In this way, the language of the heretic becomes a tool of both for the alternative-experts to claim a new expertise, based in insider status and recast with populist sentiments and outsider tropes.

Utilizing the language of the sin further establishes this heretical positionality: When discussing Dr. Malone’s deplatforming from LinkedIn, Rogan tries to understand why it happened, and Malone replies “the sin was that some one of their fact checkers...had identified the Atlantic Monthly article, attack article [which] was written about me and concluded that I was an anti-vaxxer and therefore I should not be allowed on LinkedIn” ([00:16:08 – 00:16:29]). Whereas “attack” ascribes intentionality from the mainstream to tear him down for his alleged transgressions, describing his these transgressions as a “sin” sarcastically frames the position of “anti-vax” in this establishment, religious doctrine, appealing to free-thinking sentiments that help bolster his argument while casting the mainstream in a negative light. He repeats the sin language later, referring to a group of epidemiologists who were “cancel[ed]” by the NIH and there was a supposed “concerted effort on the part of the federal government to destroy them” because of the “sin” of “raising questions about the effectiveness of vaccine lockdowns” (Malone, [00:50:35 – 00:50:56]). A supposed concerted attack seems to be an out of proportions

attack for just “asking questions,” especially for a viewing group defined by their free-thinking ideology. Thus, associating the word “sin” with “asking questions” inverts the traditional meaning of sin, and connects the mainstream government with the dogma and crusade style of the church. The sinners, then, become in the right, and the alternative-experts heretical ideas become a strength, increasing their credibility instead of detracting from it.

The alternative experts further restructure consensus under the framework of heresy. Recall that in Dr. Weinstein’s example of consensus formation with plate tectonics, he mentions that this idea was “an absolutely heretical idea when it was introduced, the idea that the continents are moving, wow, that was mind-blowing and blowing and almost nobody got it at first, right? Today, everybody gets it” (“Weinstein & Kory” [00:30:28 - 00:31:52]). The specific choice of the word “heretic,” combined with the fact that they continue to call themselves and their work heretical, implies that their claims too, will eventually be accepted, even though they face intense backlash from the dogmatic mainstream now.

The alternative-experts explicitly state they value this form of science, which is correct, and condemn the mainstream is responsible for subverting it because of profit and corruption. Their moral character and health orientation abhors what they characterize as the unscientific practices of these institutions because of the immense impact it has. Simply put, Dr. Weinstein says, “Just even the loss of one person so devastates a family that just thinking about all the people who are gonna be harmed by the fact that we’re not following the evidence and figuring where it leads” (“Weinstein & Kory” [02:50:50 - 02:51:07]). His quote portrays a similar emotional gravity, and emphasizes the lack of people, in his opinion “following the evidence” and in this corrupt scientific system. In this way, the language and framing of heresy potentially opposes a religious dogmatic characterization of mainstream science. A push against the

mainstream organization results in an outsider status, but the reproduction of the cultural imaginaries of the heretic complicates this positionality with insider status.

Not only are these new ways of knowing more rhetorically appealing because they reproduce the cultural imaginaries of the insider and thus improve expert status, but this rhetoric creates a schism, between the dogmatic science of the mainstream and the science of the alternative experts. In doing so, new scientific ways of knowing are centered: a new scientific epistemology of the alternative-experts, or an alternative-science, emerges. The next section will explore the many ways in which the scientific epistemology of the alternative-scientists, or the alternative-science, does not only reproduce biomedical scientific methods, but fundamentally reshapes what science is.

New Epistemologies: The Maverick's Science

Alternative experts don't simply reject mainstream science: they construct a parallel epistemology rooted in old science but departing from it. Returning to Meyer's exploration of Climate Change counterknowledge and counter-expertise in science related populism, we learn that we cannot merely retreat to a formulation of populist common sense rejecting evidence, as this is an oversimplification of what really happens. Instead, he argues that science-related populism "entails a pluralization of the sorts of expertise and practices from which truth might be recognized as emerging" (Meyer 856), which demonstrates how expertise and knowledge making undergo epistemic fragmentation under these ideologies. Meyer's populists do not abandon science, instead, they pursue a new form of it.

Similarly, I argue that the alternative-experts elevate new forms of knowledge, and construct an "alternative-science," by recasting insider ideals of science with populist and democratic ideals. This alternative-science is populist, because as Meyer writes, science related populism is not epistemically based "upon their separation from the common world, but by their immersion in it. This immersion is then characterized by an engagement and a caring relation to the subjects of its knowledge and action" (Meyer 857). The caring aspect speaks to the health and healing orientation I discuss throughout this thesis, merging the cultural imaginary of health and healing with a for the people, caring orientation. Similarly, while he cautions against merely seeing populism as common-sense as anti-science (857), I argue that his science related populism should be characterized not only as using caring as a basis for new epistemologies but using the common sense of the "common world" to do so. I also argue that it should be understood as democratic, as it contains democratic appeals to accessible and citizen-based knowledge production. In this way, they rhetorically posit a way of knowing that appeals to the public

imagination of what science *should* be, as recast under populist and democratic ideas— making this form of knowledge straddle both insider cultural imaginaries and outsider ideals of populism and democracy. By aligning themselves with this new form of knowing with insider imaginaries and populist ideals, they gain widespread appeal.

Anecdotes and the Primacy of Common-Sense

“The party told you to reject the evidence of your eyes and ears. It was their final, most essential command.” — George Orwell, 1984

Anecdote of clinical experience and population epidemiology function powerfully as another form of populist counter knowledge because anecdotes colorfully draw out evidence that is immersed in real-life. As a persuasion technique, anecdote is more salient than statistics in science communication when appealing to laypersons, partially due to the ease of communication and accessibility of this information but also because under populist meta-narratives, which elevate common sense. As previously discussed by Meyer, populist scientific evidence immerses itself in the “common world” instead of drawing back from it, and the fully fleshed out narratives of anecdotes trump cold, impartial statistics when it comes to being immersed in the common world. Moreover, anecdotes are typically more self-evident and accessible to digest than statistics, which elevates their value. Indeed, as Starr notes, “many Americans, epitomized by the Populists, continued to believe in the adequacy of common sense and to resist the claims of the professions” (Starr, 19). Thus anecdotes, as a tangible example of something that is witnessed, appear to be common sense: I saw it with my own eyes, so therefore it must be true.

The liberal invocation of anecdotes by the alternative-experts, conceptualized under this framework of populist immersion in the world and populist common sense, are incredibly thus appealing to laypersons. Consider Dr. Kory's statement, "I know 29-year-olds who are disabled — literally healthy 29-year-olds who can't go back to work." ("Weinstein & Kory" [02:07:14 – 02:09:00]). Similarly, vague invocations of anecdotes allow people to draw upon their own experiences. Dr. McCullough states, "they're one of the thousands of people who dropped dead within a couple days of the vaccine, ("McCullough" [02:01:14 – 02:04:42]). There is a sheer terror to these stories that is not as well conveyed by statements about percentages. Moreover, anecdotes function as common-sense evidence, which under a populist framework, have a higher appeal. In this way, anecdotes become an appealing form of evidence under populism. Moreover, though Starr argues that common sense can be used to resist the claims of the professionals, the script becomes more complicated when it is the professionals (the alternative-experts) who are using common sense appeals such as anecdotes. Though their professional, academically grounded credentials cement their insider status, their use of anecdotes allows them to place one foot in the outsider camp as well. Whereas mainstream experts rely on statistics and thus constitute a specific form of scientific evidence and embodiment (ie, the scientist or expert who uses these forms), by relying on common sense forms of evidence the alternative-experts are situated within an embodiment of populist way of knowing or populist epistemology. Their embodiment as a populist knower lends credibility to them amongst a populist-learning viewership, thus cementing their expert appeal.

Moreover, many of their anecdotes tend to be grounded within clinical encounters, creating a primacy of clinical anecdotes over general anecdotes. This introduces a subtlety, where the credentials and positionality of a physician improves the credibility of an anecdotal

way of knowing, showing in action how insider and outsider appeals are straddled in making these claims. Dr. Kory states, “I have dozens of testimonials of people who were sick for months. They took ivermectin and within 12-24 hours started to feel better” (“Weinstein & Kory” [02:07:14 – 02:08:46]). The relief of dozens of people focuses on magnitude and common sense: they took the drug, and they were cured, so therefore the drug works. Dr. McCullough similarly grounds his claims in observations from his clinical work: “I say, get monoclonal antibodies in the ER... I was relentless... I was relentless.... Thank the Lord. They got into the AstraZeneca monoclonal antibody trial. We don't know if they got placebo or monoclonals, but she survived” (“McCullough” [02:13:04 - 02:15:58]). Dr. McCullough is operating as a quasi-physician, not directly able to prescribe the medicine but essentially prescribing his recommendation for her to get into a clinical trial so she might gain access to a certain medication. Despite him acknowledging that there is only a 50% shot she was in the group that received the treatment, not the placebo, this conclusion is framed as a successful demonstration of the efficacy of this treatment. Moreover, the impassioned nature of the “relentless” action and the cry, “Thank God,” improves Dr. McCullough’s moral character by situating him as a doctor who is willing to fight for what’s right to get his patients better, despite the difficulties, which as the previous section discusses, allows him to claim expertise through an enhanced moral character. Regardless, the utilization of these anecdotes from a clinical perspective combines the common sense of a narrative form of storytelling with the credentials and academic position of a clinician, merging outsider evidence with insider status, and thus creating a paradoxical positionality which is perceived as expertise.

These anecdotes also operate at a population scale, gesturing at an anecdotal form of epidemiology without actually utilizing the mainstream-tactics of epidemiologists. Dr. Kory

states, "look at India, [which had] lots of Delta variants... Ivermectin was slaying the Delta variant," or "Zimbabwe... [which] basically eradicated COVID with the widespread adoption of Ivermectin" ("Weinstein & Kory" [1:35:15 – 01:36:49]). Seeing a country that has low levels of COVID and "widespread" use of a particular drug, ivermectin, is a powerful story which implies causation: further use of potent verbs such as "slaying" and "eradicat[ing]" COVID emphasize this. The evidence is circumstantial, and mainstream scientific tactics would zoom in closer to look at any confounding variables, yet this detail-oriented approach simply is not as appealing as the common sense conclusion that the drug fixed the problem. This also takes advantage of the magnitude heuristic and relies on numbers to make his point, because numbers constitute a language of science. He zooms in on "Uttar Pradesh... a state [in India] of 240 million people..." who he claims were using ivermectin "aggressively and they have some of the best numbers" ([2:52:12 – 2:52:35]). Later in the podcast episode, he states India, South Africa, Zimbabwe, certain cities in Brazil, UK, Slovakia, Czech Republic have all used Ivermectin and have been successful in lowering COVID rates ([1:35:15 – 1:36:49]). All of these country-level examples utilize the magnitude heuristic, as countries are very big, but invoking the number 240 million focuses on the necessity of a large sample size, and even using the word "numbers" in the quote to refer to some vague, undefined outcome relating to COVID, suggests some level of epidemiological and statistical rigor. This suggestion is made explicit, as Dr. Kory calls this evidence "epidemiological data," ([1:35:15 – 1:36:49]) despite the fact that it is likely correlational because it is based on observations, not studies. This obfuscates the anecdotal nature of these claims, demonstrating how a sophisticated rhetorical anecdote arises that appears more credible than it actually is. The sheer scale of these examples lends them a rhetorical weight that is appealing under populist ways of thinking which emphasize common sense

conclusions, even though their methodological validity is left unestablished. Indeed, in the light of this data, Dr. Weinstein states that one must, "...extrapolate, what would a reasonable person do when faced with a safe drug with noisy data that has a very strong signal of efficacy that works both as a treatment and as a prophylaxis? What would you do if you were in charge? ([01:37:01 – 01:37:48]). This rhetorical question is set up to have a (seemingly) obvious conclusion– with a “strong signal” and any methodological concerns dismissed as simply “noise,” any “reasonable” person would choose to act. This both denigrates the mainstream as unreasonable, while flattering the layperson, who is situated in the shoes of the actor as someone who acts “reasonabl[y]” because they see what the mainstream somehow does not. In this way, despite their fundamental limitations as observational and correlational, population based anecdotes become central ways of knowing to the alternative-experts. Their common sense appeal allows the alternative-experts to gain status.

Notably, the use of anecdotes does not prevent the use of statistical and mainstream methods such as clinical trials, as described in chapter 1– However, this form of common sense evidence still takes primacy under a populist framework of thought. Thus, when Dr. Weinstein says: “We do not have clinical trials to support that protocol. All we have is clinical experience,” a mainstream expert would view this as undercutting the strength of the evidence, as clinical trials are typically viewed above clinical experience. However, the primacy of the clinical anecdote within their viewership allows this statement to not actually detract from the argument.

²⁵ Thus, this is not only a dual rhetoric, of using both inside and outsider mechanisms, but actually demonstrates how the alternative-experts embrace of common sense comes to lend them

²⁵ Moreover, this acknowledgement actually gestures at Weinstein’s intellectual humility, which actually improves his expert credibility. The role of the moral character the alternative-experts present cannot be understated, due to the previously discussed influence of morals and value orientations on the credibility of an expert.

an increased appeal compared to the mainstream-experts, whose arguments are limited to more statistical and mainstream ways of knowing.

This rhetoric of common sense, evidence I witnessed with my own eyes is not only utilized to build up the evidence base of the alternative-expert, but is also used to tear down and discredit the claims of the mainstream. As epitomized by the epigraph to this section from Orwell's famous 1984, authoritarianism is linked to the denial of common sense evidence. Thus, the alternative-experts further condemn the mainstream by positing that they are denying the evidence people can see in their everyday lives. Indeed, Dr. Weinstein states "these things [facts and consensus] are being handed down from on high, and then they are silencing the people who are saying, "Hey wait a minute, that thing that you just told me from on high, doesn't square with all the stuff that I can see" ("Weinstein & Kory" [00:32:06 - 00:32:17]). This restatement of Orwell's famous quote appeals to anti-authoritarian sentiments, allowing this form of rhetoric to not only create a convincing epistemology for the alternative-experts but to condemn the mainstream-experts.

The downfall of the Randomized Control Trial (RCT) and the rise of new forms of evidence

The Randomized Control Trial (or RCT), which is often viewed as the gold-standard of evidence in mainstream biomedicine, is painted as dogmatic and against the people. The methods themselves are condemned as detached from reality and thus ineffective for the real-life concerns of everyday people (the populists primary orientation is the good of the everyday man).

Moreover, this results in harm done to the American people and the everyday individual. Indeed, returning to Meyer's supposition that a populist way of knowing is based on relations of care,

any form of evidence that is bad for the people and society as a whole is denigrated. In this way, the mainstream RCT becomes anti-populist and loses its rhetorical appeal. Engaging with populist alternative-methods, such as the anecdotes previously discussed, then becomes rhetorically salient and increases the credibility of experts.

Indeed, the RCT becomes an object described in the language of mental health, delusion and sickness. This language makes the RCT the folly of the mainstream, delegitimizing it as a valid method of knowledge production. This ultimately allows the alternative-methods of the alternative-expert to appear more persuasive and scientific, increasing their credibility. Returning to Dr. Weinstein and Dr. Kory's back and forth on the value of anecdotal, population level data, we see that the RCT is painted in stark contrast as the worst alternative to their methods. Dr Weinstein states, one must:

...extrapolate, what would a reasonable person do [when] faced with a safe drug with noisy data that has a very strong signal of efficacy that works both as a treatment and as a prophylaxis? What would you do if you were in charge? And what you hear back is the most maddening? [people saying] 'Well, you know, I'm evidence-based and if it isn't a large-scale, randomized controlled trial, then it isn't evidence to me.' And it's like only a crazy person would say that in this case, and yet you hear it all the time. ("Weinstein & Kory" [01:37:01 – 01:37:48])

Dr. Weinstein's use of population scale anecdotes (that are ultimately observational and correlational, but that appear persuasive, as previously discussed) are "reasonable," whereas the "evidence-based" methods of the RCT are "maddening" and something "only a crazy person would say." This claim about RCTs only works because it comes after the alternative-experts have built up an evidence base of their own. Thus, engagement with the RCT is painted as being done just for the sake of having the RCT, as other evidence (supposedly) exists that would allow people to take action and solve the crisis of the pandemic. Any action that does not work towards solving the pandemic, is against the common man in populist metanarratives. Dr. Kory

continues, bemoaning the what he calls the “ivory-tower syndrome, or this evidence-based maniacism, which is this obsession with this big randomized control trial... where they won't believe anything until you do that trial” (“Weinstein & Kory” [01:39:10 - 01:39:27]). Invoking the stereotype of the ivory-tower, or the disconnected academic whose work is inapplicable to reality, paints the RCT as disconnected from reality as well. The use of the language of a “syndrome” and the mad language of the obsessive maniac connects the RCT to a delusional, obsessive-like sickness. Ultimately, this type of characterization is antithetical to the sober, rational methods of science, which demonstrates how the alternative-epistemologies somehow seem to reproduce the cultural imaginaries of science better than the mainstream.

The alternative experts further emphasizes the RCT’s detachment of reality, which allows them to engage within populist metanarratives that prioritize engagement and common sense and thus gain credibility, by using colorful metaphors that make his point incredibly accessible to laypersons. For example, Dr. Weinstein states, “The insistence on large randomized controlled trials is like insisting on video documentation of a crime,” which implies other evidence can be sufficient ([02:45:00 – 02:46:26]). This point is emphasized by another metaphor, where Dr. Kory says, insisting on an RCT is like when “your house is on fire and there's a bucket of water and somebody stops you from using it because you haven't proved it's water” [01:55:4 – 01:55:49]). Alongside similarly painting the RCT as overburdenous and potentially harmful, the insistence that the mainstream refuses to believe there is water in the bucket until they have an RCT further paints the mainstream as detached from reality. It implies that the evidence we have now is sufficient to conclude that the water is in fact-water (or in the case of COVID, that ivermectin is effective) is simple common sense, making it particularly appealing rhetorically.

The urgency of the pandemic situation is further used to characterize RCTs as too limited in its capacity to help the everyday man in the here and now. As Dr. McCullough states, “We knew [there] was insufficient time for large randomized trials” which can “take two to four years... We don't have two to four years. This is a mass casualty situation. We use the precautionary principle...” (“McCullough” [00:07:13 – 00:09:17]) The dismissal of the RCT is subtle, seemingly rational: we have no time, and he decides that instead it is best to operate on the “precautionary principle,” which Dr. Weinstein summarizes as finding a “safe med that seems like it only has upside” and deciding “let's just do it.” (“Weinstein & Kory” [01:26:37 – 01:28:10]) This urgent need to help others becomes rhetorically more convincing than waiting for the slow, methodical results of a clinical trial. Furthermore, the alternative-experts state that even creating an RCT at this point would be downright unethical, furthering the apparent incompatibility between the RCT and the caring, populist epistemology. Dr. Kory states, ‘I actually think that trial is unethical, I could not as a physician, knowing what I know, give someone a placebo right now, for Ivermectin, the evidence is too overwhelming’ (“Weinstein & Kory” [01:17:43 – 01:19:09]). This moral condemnation necessarily raises the status of other forms of evidence while denigrating the RCT as uncaring, and thus unappealing in a populist light.

Indeed, the invocations of other, populist forms of evidence allow the denigration of the RCT to continue. Returning to the schism I set up while discussing the Mainstream Dogma and the Heretical Alternative-expert, I argue that this downfall allows for the methods of the alternative-experts, in contrast to the RCT to appear to be for the good of the people and more epistemologically appealing under populist ideals. Similarly invoking religious ideas, alternative experts position themselves as a Cassandra-type, discarded prophet, creating credibility and

authority that is later backed up by events that come to pass. For example, Dr. Kory discusses how his testimony in the senate is discredited then later affirmed, stating,

When I said to use corticosteroids in the Senate testimony, I was attacked, criticized. There were-- 'cause there was no randomized controlled trial. Seven weeks later, Oxford put out the recovery trial, which is their big trial in the UK and they showed the corticosteroids were life-saving. So, we were validated. (Weinstein & Kory [02:03:00 - 02:03:20])

Indeed, the Oxford recovery trial on the corticosteroids, did demonstrate that Dr. Kory's claims about corticosteroids were correct, and this drug was subsequently widely used and very effective in reducing death (The RECOVERY Collaborative Group; Mourad et al.). He is correct, and it allows him to paint the mainstream as further dogmatic. Importantly, it allows his condemned methods to appear more salient and valid: in justifying his claims about corticosteroids before the RCT was released, he states that he based his recommendations on a "group of scholars reviewed all of the trials from SARS, MERS, and H1N1, so the prior pandemics...they found was that corticosteroids were actually life-saving in the prior Coronavirus pandemics" as well as the fact that data, "Doctors, some of them anonymously were starting to post like, "Hey, we're using steroids... and we're finding [the patients are] not getting intubated...they're coming off ventilators and we're actually discharging patients," and this was "real-time feedback that it was working" ("Weinstein & Kory" [01:59:03 – 02:04:05]). This evidence centers on common-sense anecdotal clinical evidence, which as previously discussed, epitomizes real-world immersion and is therefore appealing to the populist. However, it also exposes a reliance on preliminary data and logical application of previous findings. This form of knowledge centers common sense as scientific knowledge: it worked before on similar pandemics, therefore it should work now. In this way, a populist form of evidence takes center stage in this discourse.

While the ultimate conclusion was correct in this single case, this does not mean that disregarding the RCT for other forms of evidence is necessarily wise, even though it appeals to populist common sense appeals. It only takes one anecdote to demonstrate their track record, as their detractors are not present on the podcast to show the many other times these methods and logic lead them astray. For example, consider another one of Dr. Kory's claims that Ivermectin works against COVID. He similarly invokes anecdote as well as preliminary evidence, such as the famous Australian cell culture study showing that ivermectin eradicated virus "within 48 hours" in monkey kidney cells ("Weinstein & Kory" [00:03:05 – 00:04:57]). This study was the main reason why Ivermectin became a salient tool to study for COVID. Though there was nothing wrong with the Australian study itself, the fact that later efforts to use Ivermectin to treat COVID did not work demonstrates a common scientific phenomenon— often, studies that work in a lab, on a single cell or on another organism do not have the same effect when applied to an entire human being, which is a lot more complicated than a single cell and has properties different from model organisms. This demonstrates why the RCT is ultimately necessary— while often it will concur with preliminary and anecdotal evidence, there are instances where it does not, and thus it is necessary to perform it. Yet regardless, the science of the alternative-experts rhetorically is more persuasive under populist appeals.²⁶

In the examples of anecdotes and preliminary evidence extrapolation, the RCT was denigrated not on the basis of its ability to reach accurate conclusions, but on its overburdenous and slow character when other forms of evidence exist. However, the alternative-experts also attack the science of the RCT by appealing to the scientific imaginary of consensus and

²⁶ Of course, the mainstream did successfully cut through the slowness of RCTs through coordinated efforts and by removing bureaucratic obstacles when it came to the RCTs for the vaccine, however, this point isn't rhetorically widespread, and this is not as effective as the alternative-experts claims.

communal validation.. For example, Dr. Kory discusses the hierarchy of different types of evidence in medicine, constructing it as a pyramid:

“when you look at the strength of medical evidence to prove something in medicine, you start at the bottom, which is an anecdote, right?...Then you have like observational trials where you just follow a group of patients, or you look at a group that you treated versus who you didn't maybe retrospectively....the top of that pyramid is not a large randomized controlled trial. It's actually what Bret said. It's a meta-analysis of randomized controlled trials” (“Weinstein & Kory” [00:46:04 –00:48:58])

His assertion that “the top of that pyramid is not a large randomized control trial” implies that while many other experts may treat it this way, his community of alternative-experts understand the limitations in this form of evidence generation better than the mainstream-experts. He explains why it is flawed in a convincing way: “We all like big randomized controlled trials when you can get them, even though they-they are prone to error” because RCTs are individual studies, and “any individual trial can have an error, or a flaw, or a dosing, or a timing problem, it might lead you to the wrong conclusion” ([00:46:04 –00:48:58]). He does not completely discount them, as demonstrated by his point that “we all like” them “when you can get them,” but makes the case for how individual trials can be flawed. Condemning this form of evidence allows the the meta-analysis to be situated as a better form of evidence in this community: he devised a meta-analysis as when you take a “whole collection of [randomized control] trials and then you put them all together and you look for the signal out of that, it's much more robust because it corrects for any individual flaws that you'll see in studies” ([00:46:04 –00:48:58]). In other words, it is the same argument as looking for a consensus: if amongst multiple trials, the signal points to a conclusion, the outcome is more reliable than any one trial, which may be flawed. This reproduces the cultural imaginaries of science but reaches a slightly different conclusion than the mainstream as per methodology. It is also an appeal to common sense: summed up as more trials are better than one. In this way is a very subtle

recasting, as meta-analysis are actually highly valued in biomedical science alongside RCT's, but where Dr. Kory points out the limitations of RCTs and touts meta-analysis as a solution and introduces no flaws, mainstream science recognizes the use of both and the flaws of both.

Comically, Dr. Kory himself demonstrates why the mainstream does not value meta-analysis over randomized control trials, but instead sees both equally as valid forms of evidence that can produce flaws. Dr. Kory concludes that “when we talk about that there are meta-analyses of randomized control trials, 24 randomized control trials, thousands of patients, that's fairly unassailable evidence” ([00:46:04 –00:48:58]). Though this invocation of the number 24 seems somewhat random in the context of the episode itself, further research reveals that Dr. Kory's own meta-analysis on a hospital protocol for COVID (the MATH+ protocol, which was retracted) uses 24 studies. This meta-analysis was retracted because 2 of his 24 included RCTs were themselves either retracted or received allegations of fraud, and later analysis by another scientist shows that when those two studies were removed, the results of the meta-analysis became statistically insignificant, which was because the effect sizes of the two fraudulent studies were so large that they profoundly impacted the study results. Dr. Kory's own work provides the evidence for why meta-analyses do not always resolve individual studies.

The rhetoric of the alternative-experts allows them to center other forms of reasoning and evidence such as the precautionary principle and the meta-analysis. Their ability to elevate these new mechanisms of knowledge making which the mainstream would not claim, is part of their outside status. Yet simultaneously, part of the appeal of these methods is in the cultural imaginaries of what science should be, and their recasting of these appeals under populist ways of knowing. It is important to note that the alternative-experts have a motivating factor which causes them to abandon RCT's: they typically do not show efficacy of their intervention of

choice. Regardless of the intent, the effect is the same: RCT's become denigrated as the gold standard of evidence within the alternative-scientific community, leaving room for other forms of rationalization and evidence, such as meta-analysis, to take their place amongst these viewers.

However, it is not only the recasting of science through populist appeals, but democratic ones as well.

Democratized Science in the Public Domain

This alternative-science does not only produce different knowledge, it produces that knowledge in different places. Aside from utilizing populist, common sense-forms of evidence, the utilization of alternative social media and emphasis on direct communication with the layperson functions reflect populist and democratic ideals of transparency, accountability, and lay involvement in processes. This spatial disparity between alternative-expert and mainstream expert science production stems from the role of the maverick: As Goldenberg notes, "the maverick's work takes place on the public stage" (Goldenberg, 159), and the alternative-experts embody this fully: where mainstream science operates through peer review, academic conferences, and institutional gatekeeping, the alternative-experts situate their discourse on podcasts, substacks, and social media platforms and in other forums which promote public discourse. The choice of venue is itself a rhetorical act, as this new form of science appeals with democratic and populist values.

The prevalence of social media as a platform for these experts gestures towards their epistemological populism. Joe Rogan's show is hosted on youtube and social media, and these media outlets are inherently populist— they differ from traditional media in the fact that they give anyone the opportunity to have a voice, no matter their training, and they open up accessible venues for citizen-journalism and commentary on society. Even outside of the Joe Rogan experience, these guests are incredibly engaged with social media: Dr. Weinstein hosts his own Dark Horse podcast on youtube, and Dr. Malone has a substack, for example. Their willingness to use these sources gestures towards their populist embodiment. They frame the move to public platforms as a democratic corrective to the failures of mainstream institutions. As Weinstein states, "podcasts would be a natural place for doctors and scientists to get together and say, what

we're seeing doesn't add up.” (“Weinstein & Kory” [00:12:53 – 00:13:06]) this positions podcasts not as an informal medium but as a legitimate venue for scientific discourse— one that is accessible to all, in a way that gated journals, conferences, and mainstream academic mechanisms of knowledge production are not. Despite this, another statement from Dr. Kory seemingly confirms this construction of social media as a venue for scientific community building and collaboration (especially in the face of corruption): Dr. Kory says, “It started popping up... doctors, some of them anonymously, posting: ‘Hey, we’re using steroids and we’re finding them not getting intubated’” ([01:59:03 – 02:04:05]). In this way Social media also functions as a venue for a distributed, decentralized form of scientific community-building which mirrors the communal validation discussed in Chapter 1, but at a broader scale and in a more democratic light. The public visibility of the social media approach encourages a democratic involvement and participation that the mainstream forums do not, allowing the alternative experts to demonstrate their reliance on populist ways of knowing.

They similarly condemn elite institutions of academia for their closed door approach, and call for more public democratic involvement in science. Dr. McCullough tells Rogan he hosts public talks with other physicians reviewing COVID data, akin to “medical grant rounds for the public. And when I tell people I said, where are the medical schools doing this? How come the medical schools aren't having public symposiums?...Why are there no review of the data?” (“McCullough” [02:23:56 - 02:26:33]) Mainstream-experts would argue that there are reviews of the data, in scientific journals and other biomedical forums, but Dr. McCullough is positing that reviews of the data should be public. This further speaks to the alternative-experts values of transparency and democratic accountability. Because their values mirror the values of the listener, their credibility is increased and thus their expertise is recognized.

"Do Your Own Research" as Epistemology, power to the people

Finally, a populist and democratic way of knowing centers the role of the layperson in forming scientific knowledge. Though mainstream science views it as the position of scientists, an implication of using common-sense evidence is that laypersons can observe on their own what is happening and draw their own conclusions. The imperative to *Do Your Own Research* is common in vaccine hesitant discourse (Chinn et al.), and by encouraging this relocation of epistemological power, the alternative-exerts flatter the layperson and align with their values, increasing their credibility and thus their own expert status.

The alternative experts' eschew mainstream science for common sense, positioning the layperson as a scientist of common sense: As Dr. Kory states, "When you have a super spreader event like choir practice and 59 people come home with COVID while socially distancing— you don't need to be a high-level scientist to know that was probably airborne transmission." ("Weinstein & Kory" [00:36:19 – 00:38:09]). The implication that you do not need to be a scientist is particularly damning, because he is responding to what he describes as the slow movement and conclusions of mainstream-experts in realizing COVID was airborne: as he says, "the officialdom was that it wasn't airborne. So, like, it was basic stuff that didn't make sense." So, this quote serves a dual purpose of setting up the juxtaposition between the corrupt mainstream, who are failing to deliver what is characterized as seemingly obvious and timely conclusions, and the common-sense of the everyday layperson. Dr. Kory furthers his point about the seeming contraction when "something [scientifically] is really, really clear, and yet it doesn't comport with what's coming down" ("Weinstein & Kory" [00:29:50 – 00:38:09]). The use of "coming down" draws out a perceived power imbalance, further emphasizing the distance between the corrupt, mainstream elite and the people as they purportedly twist reality.

Moreover, this creates an expert who does not demand deference: by centering the role of the layperson as their own thinker and knower, they center the values of free-thinking which will appeal to Rogan's viewership while appealing to an idea as an expert as empowering laypersons to use their own knowledge. As Starr writes, "authority signifies a potential to use force or persuasion, though paradoxically authority ends when either of these is openly employed (Starr 9). Thus, any persuasion must seem minimal for the alternative-experts' own credibility to thrive. The alternative-experts, by centering the agency of the layperson in making medical decisions, do not appear to be exerting authority, which paradoxically enhances this authority. Malone states "my position all the way through this comes off of the platform of bioethics and the importance of informed consent... to appropriately choose to participate in a medical experiment they have to be fully informed of the risks as well as the benefits and so I've tried really hard to make sure that people have access to the information about those risks and potential benefits the true unfiltered academic papers and raw data etc..." ("Malone" [00:18:35 – 00:18:42]). He positions the data and papers as units of truth, and himself merely a messenger, who does not alter the message by presenting it. He places the layperson in the role of the scientist, and fades into the background. Social Constructionists would point out that there is no such thing as an "unfiltered" paper, all scientific work is subject to interpretation,²⁷ and anyone who has read multiple contrasting *New York Times* about the benefits/ harms of one glass of wine a night will know that presenting "raw" data can lead to a multiplicity of conclusions, some better than others, yet this rhetorical move is still appealing. As he fades into the background, any persuasive appeals become invisible, and thus his authority is valid. Moreover, this rhetoric

²⁷ See Mitchell's, "The Fault in his seeds" for a discussion on how Paul Morton, infamous racist phrenologist, and anatomist Friedrich Tiedemann, abolitionist, both operated with similar data sets for brain size and yet reached wildly different conclusions– the former used the work to support racist theories about the superiority of white people, the latter used it to argue there weren't biological differences between races (Mitchell).

centers the layperson as the scientist and the final decision maker, centering layperson values of autonomy and agency, which makes Malone more appealing. The mainstream, by contrast, is undercut, as not offering “informed consent” or centering the agency of laypersons during what Malone calls an “experiment,” effectively undercutting the evidence and work supporting mainstream interventions. Finally, the invocation of “bioethics” as shaping Malone allows him to present himself as a moral, and thus credible figure.

Centering the layperson’s decision making capacity is incredibly flattering, which furthers the appeal to the layperson as a decision maker: While discussing different countries and their relative successes and failures in managing COVID, alongside their demographic data, Dr. Malone states, "You can look it up on Worldometer — you don't have to believe me, your audience is smart enough" (“Malone” [00:56:33 – 00:56:52]). Of course, the specific part that can be looked up on the worldometer—specific details about population data— is a small part of Malone’s overall argument, yet it positions this information as a simple google search away, and limits the visibility of Malone’s own interpretations of the data he is presenting. It is particularly appealing in contrast to the mainstream, which is painted as doing the opposite in trying to suppress the epistemological efforts of the layperson: Malone cites a “New York Times [article with an] epidemiologist and a vaccinologist and the title was how you should think about children and Omicron. It was blatantly saying "this is how you should think- we're going to tell you how to think” (“Malone” [02:38:25 – 02:38:50]). This rhetoric is also used when it comes to non-scientific speculation to avoid making incredibly controversial claims: Malone discusses a meeting that took place between someone in the Biden administration and Indian Prime Minister Modi, while Uttar Pradesh was supposedly thriving after widespread adoption of Ivermectin to treat COVID. While Malone states “all I know is that immediately afterward there was a decision

not to disclose the contents of what was being deployed in Uttar Pradesh,” and lets Rogan speculate as to how “nuts” it is that somewhere that was thriving during COVID wouldn’t share their information, all Malone states is “here are the facts here are the verifiable data draw your own conclusion” (“Malone” [00:33:01 – 00:35:00]). It is unclear what this meeting was, if it happened, or if it actually discussed COVID or not, but Malone suggests a conclusion to Rogan, the layperson stand-in, which he infers, instead of directly stating it. This both provides the layperson with a sense of involvement and decision making that is appealing under populist frameworks, while allowing plausible deniability for Malone.

This charge isn’t entirely unfair, as mainstream science often fails to be fully transparent as to how they reach their conclusions when it comes to science communication, and public health has paternalistic aspects. The critique is reasonable, and mainstream-science needs to take an active role in rebuilding trust. However, until that point, the alternative-experts are able to appeal to this critique to fully discredit the broader enterprise of mainstream institutional science and expertise, replacing it with an epistemology where individual layperson judgement and common sense become elevated and the alternative-expert’s support of these values serves to support their own expert status.

The Faux Outsider: Conclusion

Expertise emerges by simultaneously positioning oneself at the top of a *constructed* scientific body while establishing willingness to dissent from a corrupt establishment. This is a challenge to Goldenberg's "flipped hierarchy" — it's not simply that the maverick inverts the credibility structure, but that they occupy both positions at once. This demonstrates once again how this issue is rhetorical, not ontological. This speaks to the need for mainstream institutions

to inculcate trust by creating strong legislation and penalties for profit-driven corruption, as it is the mainstream which creates the opportunity for the alternative'expert's rhetoric to flourish in the first place.

Conclusion

The rise of alternative-expertise during the COVID-19 pandemic is not an anomaly. It reflects a deep epistemological fracturing, a mistrust in mainstream institutions and a rise in populist sentiments that reshape neither an anomaly nor a passing cultural moment. It is the product of deep epistemological fractures in American public life that the pandemic brought to the forefront. This thesis has argued that alternative-experts like Dr. Robert Malone, Dr. Peter McCullough, Dr. Pierre Kory, and Dr. Bret Weinstein did not succeed by rejecting the authority of biomedical expertise, but instead by reproducing and recasting it.

Chapter One, "The Inside Man," demonstrated that the alternative-experts construct credibility by reproducing the cultural imaginaries of biomedical insider status. Drawing on Pierre Bourdieu's theory of symbolic capital, the chapter showed how the invocation of the cultural imaginaries of biomedical expertise-- academic degrees, prestigious institutional affiliations, peer-reviewed publications, and clinical anecdotes functions not merely as biographical information but as a currency through which laypersons recognize and confer expertise. Their accumulated credentials, from Dr. Malone's invocations of Harvard and the Salk Institute to Dr. McCullough's hundreds of published papers and Dr. Kory's ICU practice, situate these figures within the consensus based, communal structure of the sciences, allowing these figures to further gain recognition as insiders of biomedical expertise. On the other hand, the failure of mainstream-expert Dr. Sanjay Gupta to successfully reproduce these same cultural imaginaries on Rogan's podcast confirm that expertise is ultimately rhetorical. It is not enough to be an expert, one must rhetorically perform expertise to be viewed with authority.

Chapter Two, "The (Faux) Outsider," returns to the traditional cultural imaginaries of insider expertise but recasts them under new appeals as well. Whereas the maverick archetype described by Maya Goldenberg suggests that alternative-experts gain credibility through simple opposition to the establishment, this chapter argued for a more nuanced understanding: these figures do not simply flip the hierarchy of expertise but carefully recast what it means to be an insider in the first place. Through a transcendent distance of maintaining insider credentials while narrating an institutional exodus, they construct a dual positionality that is uniquely persuasive. The delegitimization of mainstream institutions as corrupt and profit-driven, a populist-democratic narrative, and the positioning of the alternative-expert as uncorrupted by financial motive and oriented solely toward the health of the American people, a potent cultural imaginary of biomedical expertise, these insider roles are combined and forge a new form of expertise. Moreover, a new way of knowing emerges from it. The alternative-experts construct an alternative, populist epistemology, similarly based in the insider epistemologies of science but deviating slightly through elevating meta-analysis over the randomized control trial, centering clinical anecdote and population anecdotes as common-sense evidence, and democratizing the sites of knowledge production through social media and a "power to the people" claim. This rhetoric further cements their credibility within an audience predisposed toward skepticism of elite institutions, but who simultaneously draw from the insider imaginaries of biomedical science.

Taken together, these two chapters reveal that alternative-experts generate credibility by reproducing and recasting the cultural imaginaries of insider biomedical expertise. Expertise is ultimately not ontological, but rhetorical, and the potency of an actor's persuasive appeal is what imbues the putative biomedical expert (either mainstream or alternative) with their authority.

The stakes of this argument have only grown in the years since the podcast episodes examined were recorded and aired. The figures at the center of this analysis have moved toward its center: Dr. Robert Malone, who is described in this thesis's opening pages as the physician whose elaborate credential-listing prompted 270 health professionals to write an open letter to Spotify, was appointed by Robert F. Kennedy Jr. to the Advisory Committee on Immunization Practices (ACIP), the governmental body that makes national vaccine recommendations. Kennedy himself, whose long history of vaccine skepticism and legal challenges to vaccine mandates exemplifies the alternative-expert archetype at the political level, was confirmed as Secretary of the Department of Health and Human Services in 2025. The construction of alternative-expertise cannot be ignored as a fringe position anymore, it is now the predominant force within our institutions of health: in a way, it threatens to subsume the mainstream all together and become it.

This proliferation of alternative-expertise, then, is a broader problem with significance outside of just this case study, and which must be addressed. The epistemological fragmentation that Baier and Lyotard theorize, in which distinct epistemic spheres generate their own experts and their own forms of evidence, has a continued potency in our era. The algorithmic architecture of social media continues to reinforce ideologically homogeneous information environments, and the continued-influence effect documented by Lewandowsky and colleagues ensures that first impressions, however misleading, outlast subsequent debunking or corrections. The alternative-experts' rhetorical strategies, examined here in a discrete moment in late 2021, remain as effective today as they were then, and have become translated into institutional power, emphasizing the impact of these alternative-experts.

The mainstream scientific community must instead internalize the rhetorical lesson of this analysis: that expertise is communicated, not merely possessed. The example of Dr. Gupta demonstrates that mainstream-experts cannot simply assume their authority will be recognized. Mainstream-experts must understand how credibility is constructed within different epistemic communities and orient their science communication efforts accordingly. This means attending not only to the accuracy of their claims but to become aware of the rhetorical dimensions of how those claims are delivered, and situate scientific knowledge within both accessible and appealing narratives.

Mainstream-experts must also grapple with the conditions that make alternative-expertise persuasive in the first place. The critique of Big Pharma that the alternative-experts deploy deeply resonates with the public because mistrust of it is rational: the documented history of pharmaceutical industry corruption, ghostwritten studies, and regulatory capture provides the foundation for their indictment. Rebuilding public trust in mainstream biomedical institutions, thus, requires not just better messaging but accountability and transparency.

Ultimately, this thesis demonstrates that the battle over COVID vaccine expertise was never simply a battle over facts. It was a battle over who gets to be an expert, who we trust, and why. As the rhetorical supersedes the ontological, understanding the rhetorical construction of alternative-expertise is the necessary first step toward reclaiming the authoritative power of biomedical expertise. And as the careers of Dr. Malone, Kennedy, and their peers have demonstrated, the stakes could not be higher.

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